NOVA Information Management School

Universidade Nova de Lisboa

Asset-Centric Temporal Graphs for Crude Oil Logistics: Schema Design and Consistency Guarantees for Multi-Resolution Network Data

Master Thesis in Information Management Systems

Pedro Porfírio

Student Number: 20241283

Supervisor: Professor Flávio Pinheiro

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Abstract

Crude oil logistics networks pose a representational challenge that existing modelling tools still tend to address through ad hoc reconciliation. Public data is published at multiple levels of resolution by overlapping agencies, reporting boundaries often fail to match operational ones, some flows are directly observed while others can only be inferred from aggregate constraints, and the underlying topology changes gradually through real operational developments that any useful representation must accommodate without constant redesign. In practice, the workflow connecting public data to downstream models usually takes the form of a custom Excel workbook or pandas notebook with hard-coded allocation rules and a residual adjustment term absorbing whatever does not balance. Although this approach can produce useful numbers, it is difficult to audit, inconsistent across analysts, and hard to extend.

This thesis proposes, implements, and validates an asset-centric temporal graph schema designed to address that gap. The schema is built around twelve architectural principles. Each physical asset is represented as a node with its own inventory and mass balance. The topology remains stable, with zero-flow edges representing inactivity rather than absence. Every node follows a universal mass balance equation with an explicit balancing item. Edges and aggregation links are derived views over a single variables table, which serves as the only source of truth. The persistent asset graph is kept separate from scenario-specific data assignments. The physical layer is distinguished structurally from the observational aggregation layer, and geography metadata is kept separate from the resolution hierarchy. Single attribution is enforced at schema level through a CHECK constraint. Flows that exist structurally but are not directly observed are declared latent rather than estimated heuristically. Bidirectional flows are represented as separate directed edges, sum-type formulas are canonical, and latent values are treated differently from unresolved ones so that source ambiguity is never hidden.

The framework is implemented as a working system composed of a relational database, a deterministic resolver that produces a per-(scenario, variable, observation date) value table, a layered view structure for downstream consumers, and a complete instance covering the U.S. crude oil network from January 2015 to December 2024. This instance includes 240 assets, 409 directed flow edges, and 1,830 variables, of which 80 are time-series-observed under strict one-to-one attribution. The resolver runs in about 20 seconds and leaves no variables unresolved. Partition closure holds within rounding tolerance across all PADD aggregates and at the USA level, and mass balance is satisfied at every node. The gap between observed and closure-derived balancing items remains small in normal periods and rises sharply around well-known disruption events, including Hurricane Harvey, COVID-19, the Colonial Pipeline shutdown, and the Strategic Petroleum Reserve release.

To demonstrate downstream use, this thesis develops a linear programme that consumes the framework's output directly. The LP takes the resolved data tensor as input, treats latent variables as decision variables, encodes the mass balance equations as equality constraints,

and represents capacity limits as inequality constraints. Applied to the Permian fan-out problem, it solves in under a second and produces a meaningful allocation of flows. This works because the framework provides exactly the upstream consistency conditions that non-trivial optimisation depends on: single attribution, partition closure, and a schema-enforced single source of truth. The thesis concludes by arguing that the same data substrate can support graph neural network forecasting models without requiring schema changes, and it outlines that development path, together with other extensions, as future work.

Keywords: logistics networks, graph databases, crude oil, mass balance, multi-resolution data, schema design, linear programming, asset-centric representation

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I am also indebted to the practitioners in the crude oil trading and analytics community whose workflows helped define the problem addressed in this thesis. The conventions established by the EIA, the IEA, and the wider public-data ecosystem reflect many years of careful work, and treating that material with the seriousness it deserves remained an important commitment throughout this project.

Finally, I would like to thank my friends and family for their patience during the final stages of writing, and my colleagues at NOVA IMS for the conversations that shaped this project in ways that only became fully clear over time.

1. Introduction

1.1 Background and motivation

Crude oil is one of the most volatile commodities in global markets, but it is also one of the most closely analysed. Producers, governments, refiners, and traders interact through spot cargoes, term contracts, and standardised futures linked to multiple regional benchmarks. Across all of these instruments, prices ultimately rest on the physical balance between supply and demand for crude itself. For that reason, any serious quantitative work on energy markets begins with an understanding of the physical system that produces, moves, stores, and consumes the commodity.

What makes this system difficult to model is the limited speed at which the physical layer can react to short-term economic signals. Well output is constrained by geology and reservoir conditions, so large and rapid production changes are technically difficult and operationally expensive (Fattouh, 2011; Ribas, Hamacher, and Street, 2010). Refineries face similar inertia: they are built for particular throughput ranges and crude slates, and moving beyond those ranges can reduce efficiency and affect product quality (Fernandes, Relvas, and Barbosa-Póvoa, 2013; Lima, Relvas, and Barbosa-Póvoa, 2016). Even changing crude grade is rarely a simple commercial choice, since it may require unit reconfiguration and planned downtime, with immediate revenue consequences. Transport introduces further delay, as pipelines, ports, and tankers impose transit times measured in days or even months (Stopford, 2009; Kazemi and Szmerekovsky, 2015). In practical terms, meaningful changes in flow usually require lead time, capital, and coordination across multiple parts of the chain (Lima, Relvas, and Barbosa-Póvoa, 2016).

Because production and refining cannot adjust quickly, inventories absorb much of the system's short-term imbalance. Tank farms, storage terminals, and strategic petroleum reserves buffer the difference between what is produced and what is consumed at any given moment. When stocks are comfortable, shocks can be absorbed with limited price impact. When stocks become tight, concerns about feedstock availability push prices upward; when they build beyond normal operating levels, storage costs rise and prices generally soften. Storage capacity also imposes a hard physical limit: once usable storage is effectively full, the marginal barrel has nowhere to go, and price formation can break down. The April 2020 WTI front-month settlement at minus 37 USD per barrel provided a public demonstration of that constraint.

Within this physical system, each operational asset—whether an onshore well, offshore platform, basin gathering network, pipeline segment, hub terminal, refinery, or export terminal—generates multivariate data that reflects its operating state. Production rates, consumption volumes, inflows and outflows, storage capacities, and inventory levels all change over time under the influence of internal constraints such as maintenance, equipment limits, and contractual obligations, as well as external shocks such as geopolitics, demand

shifts, weather, and price movements. The interdependencies across the network are nonlinear and time-varying. A disruption at an upstream production site does not remain local; it propagates through downstream pipelines and storage nodes with lags that depend on transit times, routing constraints, and operational priorities.

The main tools used to study this type of system fall into two broad families. On one side are operations research methods—such as linear programming, mixed-integer programming, and stochastic optimisation—used for strategic network design and tactical planning. These approaches are mathematically rigorous, but they usually operate on simplified abstractions that leave out much of the operational detail present in the physical system. On the other side are classical time-series forecasting methods, including ARIMA, VAR, and more recent neural approaches, which are typically applied to individual variables such as refinery throughput, regional inventories, or benchmark prices. In most cases, those variables are modelled separately from the network structure that links them to the rest of the system.

A third line of research has emerged over the past decade at the intersection of these two traditions. Graph neural networks and graph representation learning offer a way to model systems with rich relational structure, and recent surveys (Wu et al., 2020; Jin et al., 2024) describe a fast-growing literature applying these methods to traffic networks, power systems, and, increasingly, supply chains (Kosasih and Brintrup, 2022; Wasi et al., 2024; Ahn et al., 2024). What makes these methods attractive is their ability to combine two things that classical OR and standard time-series approaches often treat separately: the network structure itself and the temporal behaviour of the variables attached to it.

All three of these research streams encounter the same upstream problem before modelling can begin: how the network itself should be represented. In the crude oil case, that problem remains inadequately addressed in the existing literature. Public data is released at multiple resolutions by overlapping agencies, reporting boundaries do not align neatly with operational ones, some flows are measured directly while others can only be inferred from aggregate constraints, inventory reporting often closes only at coarser scales, and the topology changes slowly through real operational events such as pipeline reversals and terminal commissioning. A representation that could handle these features cleanly would be valuable not only for LP modellers and forecasters, but also for analysts working in the practitioner workflow that currently bridges public data and downstream modelling by hand.

1.2 Problem statement

Despite the maturity of the three research streams outlined above, there is still no widely adopted representation of crude oil logistics networks that satisfies the requirements imposed by downstream users. Existing representations do not admit multi-resolution data cleanly without ad hoc reconciliation, do not enforce single attribution at schema level, do not preserve the provenance of derived quantities through the full resolution chain, and do not clearly

separate the slowly changing physical topology from the faster-changing data assignments attached to it.

The practitioner workflow that fills this gap is familiar to anyone who has worked with public energy data. Industry analysts typically maintain Excel workbooks or pandas notebooks in which production, consumption, flows, and stocks for each PADD are loaded from EIA series, balanced through hard-coded allocation rules, and reconciled with a residual adjustment term that absorbs whatever does not add up. The workflow can produce useful numbers, and those numbers are often cited in daily industry commentary, but it does so by hiding inconsistency inside the adjustment term. It also does not generalise well: each analyst's workbook is idiosyncratic, each reflects its own assumptions, and none can be passed directly into an optimisation model or machine-learning pipeline without further manual cleaning.

The central problem this thesis addresses is therefore: **how can the crude oil logistics network be represented in a way that admits multi-resolution data, enforces single-attribution structurally, preserves provenance through derivations, and is consumable by both classical optimisation models (linear programmes) and modern machine learning models (graph neural networks) without further hand-cleaning?**

This thesis answers that question by proposing, implementing, and validating an asset-centric temporal graph schema organised around twelve architectural principles, together with a deterministic resolver that produces a single per-(scenario, variable, observation date) value table that downstream consumers can read directly.

1.3 Research objectives

The research pursues three objectives:

To **design** a graph schema that represents the crude oil logistics network at the level of physical assets, captures both structural relationships and time-series dynamics, enforces single-attribution and partition closure as schema-level guarantees, and accommodates multi-resolution public data without introducing reconciliation residuals as primary data;

To **implement** the schema as a working system, including a database representation, a deterministic resolver that produces the per-(variable, date) data tensor, a layered view structure for downstream consumers, and a complete starter instance covering the U.S. crude oil network from 2015 onwards;

To **validate** the framework by formulating linear programmes that consume its output, demonstrating that the framework's consistency properties are sufficient for non-trivial optimisation tasks, and characterising the readiness of the framework for further downstream consumers (graph neural networks for forecasting) that are left for future work.

1.4 Scope and delimitations

This research focuses on the representational framework and its first downstream consumer. The asset graph is constructed at the level of named physical assets where public data permits (with PADD-level residuals for un-named tail production), at U.S. national geographic scope, with monthly temporal granularity, and with crude oil as the only commodity grade in the initial implementation. The schema is designed for extension along all three of these dimensions (to refined products, to other countries, to finer temporal granularity), and the extension paths are described where relevant, but the implemented instance is the U.S. crude oil monthly graph from January 2015 to December 2024 inclusive.

Two important areas of work are explicitly out of scope and are positioned as future work in Chapter 10. First, forecasting. The earlier framing of this thesis envisaged a temporal graph neural network as the central modelling contribution, with the schema as supporting infrastructure. The current scope inverts this: the schema is the contribution, and forecasting is a downstream consumer that the framework is designed to support but does not itself implement. Chapter 10 develops the argument that the framework is ready for forecasting work and characterises what such work would entail. Second, real-world deployment in a production setting. The implementation is a working research artefact, not a productionised system, and questions of data pipeline reliability, operator user interfaces, monitoring, and access control are deferred.

1.5 Thesis structure

The thesis is organised in three parts. Chapters 1 through 3 set up the problem: this introduction, a focused literature review (Chapter 2) covering the three streams above and identifying the specific gap, and a domain background chapter (Chapter 3) on the U.S. crude oil network that grounds the abstract design choices in physical reality.

Chapters 4 through 8 develop the contribution: the design principles (Chapter 4) that govern the representation, the schema and construction process (Chapter 5) that gives effect to those principles, the resolver (Chapter 6) that consumes the schema and produces the per-(variable, date) data tensor, the consistency guarantees (Chapter 7) that the framework supports and the audits that demonstrate them, and the linear programme (Chapter 8) that consumes the resolved data as the first downstream application.

Chapters 9 and 10 close: a case study (Chapter 9) that walks the framework end-to-end on a concrete scenario, and a conclusion (Chapter 10) that summarises the contribution, characterises the framework's readiness for graph neural network forecasting as the natural next consumer, and outlines the development roadmap beyond the current state.

Annexes provide reference material: a primer on graph neural networks for readers who want to follow the GNN-readiness argument of Chapter 10 in detail (Annex A), a comparative discussion of asset-centric versus location-centric graph representations (Annex B), a reference for the resolver's dispatch rules (Annex C), and a catalogue of the variables in the starter scenario (Annex D).

2. Literature Review

The thesis sits at the intersection of three research streams: petroleum supply chain optimisation, which has produced the dominant tools for working with crude oil logistics at the operational level; graph data models for commodity networks and the wider graph representation learning literature, which have produced the methods on which downstream forecasting consumers depend; and energy statistics conventions, which describe how the underlying public data is published and consumed in industry practice. This chapter reviews each stream with a focus on what existing work does well and what it leaves unresolved, then synthesises the gap that motivates the present contribution.

2.1 Petroleum supply chain optimisation

The supply chain literature on crude oil and refined products is mature, with most contributions falling into one of two methodological families: deterministic mathematical programming (mixed-integer linear programming and its variants) and uncertainty-aware extensions (stochastic and robust optimisation).

Fernandes, Relvas, and Barbosa-Póvoa (2013) develop a deterministic MILP for the strategic design of multi-entity, multi-echelon, multi-product downstream petroleum supply chains, applied to the Portuguese network. Their formulation captures the strategic decisions about facility location and capacity that distinguish supply chain design from operational planning. Kazemi and Szmerekovsky (2015) extend the formulation to multi-modal transportation, demonstrating that incorporating pipeline, rail, marine, and truck modes simultaneously in the strategic decision changes the optimum substantially compared with single-mode formulations. Ghaithan, Attia, and Duffuaa (2017) construct a multi-objective optimisation model for tactical planning of integrated oil and gas supply chains, balancing cost minimisation, revenue maximisation, and service-level targets across a Saudi Arabian case study.

On the uncertainty front, Ribas, Hamacher, and Street (2010) develop stochastic programming models for integrated oil chain planning under demand, production, and price uncertainty, while Lima, Relvas, and Barbosa-Póvoa (2016) provide a comprehensive review of downstream oil supply chain management highlighting the prevalence of robust and stochastic optimisation approaches in the field. The review identifies a persistent challenge across the literature: the gap between the level of detail at which the models are formulated and the level of detail at which input data is actually available in published sources.

Common to all of these contributions is the modelling abstraction at which the network is represented. Nodes are typically large aggregates — refineries, demand centres, production regions — and edges are abstract connections weighted by cost, capacity, and lead time. The representations are designed for the strategic and tactical decisions the models address (where to locate a new refinery, how to allocate production across competing demand

centres), not for the operational granularity at which data on the system is actually published. The representations are therefore well-suited to the questions they answer and ill-suited to the question of how to represent the system itself in a way that admits multi-resolution data and supports diverse downstream consumers.

A subtler concern is that this literature treats supply chain modelling and logistics modelling as effectively interchangeable. There is a real distinction. Supply chain optimisation in its classical form addresses multi-firm coordination, procurement, facility location, and strategic capacity decisions — the questions of how the network should be designed and which firms should make which commitments. Logistics network modelling, by contrast, addresses the operation of a given network under a given design: how flows actually move through fixed assets, where inventory accumulates, what consistency the public data on these flows actually exhibits. The thesis is squarely in the logistics tradition, and the distinction is worth marking because the design choices that follow (asset-centric nodes, fixed topology, schema-level consistency) reflect logistics requirements rather than supply chain ones.

2.2 Graph data models for commodity and logistics networks

The application of graph data models to commodity networks specifically — as opposed to graph methods applied to supply chains as a whole — is relatively recent. Kosasih and Brintrup (2022) demonstrate that GNN-based methods can predict hidden links in supply chain networks, identifying latent dependencies that aggregate-level data misses. Wasi et al. (2024) introduce the SupplyGraph benchmark, the first dataset specifically designed for GNN-based supply chain analytics, with explicit benchmarks for demand forecasting, production planning, and product classification. Ahn et al. (2024) develop a GNN-based probabilistic model for supply and inventory prediction using data from a global consumer goods company, demonstrating that attention-based graph architectures improve prediction accuracy by capturing cascading effects across network nodes.

These contributions are important because they validate the proposition that graph methods add value over node-level time-series methods in commodity contexts. They also share a common limitation from the perspective of this thesis: each defines its own ad-hoc data representation, and the representations are tied to the specific application the paper addresses. The SupplyGraph dataset, for instance, is structured around production lines and demand centres in a fast-moving consumer goods setting; its node taxonomy, edge semantics, and time granularity are appropriate for that setting and would not transfer cleanly to crude oil logistics.

A broader literature in graph representation learning provides the methodological backdrop. Foundational work on random-walk embeddings (Perozzi et al., 2014; Grover and Leskovec, 2016; Tang et al., 2015) and on graph neural networks proper (Kipf and Welling, 2017; Veličković et al., 2018; Hamilton, Ying, and Leskovec, 2017) established the methods that more recent work builds on. Surveys by Wu et al. (2020) and Jin et al. (2024) organise this

rapidly expanding field, the latter specifically at the intersection with time-series analysis. Temporal extensions (Xu et al., 2020 for TGAT; Rossi et al., 2020 for TGN; Wang et al., 2021 for causal anonymous walks) and hybrid architectures combining graph methods with sequence models (Li et al., 2018 for DCRNN; Yu, Yin, and Zhu, 2018 for STGCN) handle the time dimension explicitly.

Of particular relevance to the present work is the traffic forecasting literature, in which DCRNN and STGCN have become canonical baselines. Traffic networks share several structural properties with crude oil logistics networks: a stable physical topology, fully characterised nodes (every traffic sensor is identified, located, and equipped with the same set of variables), and a forecasting target (future traffic flow) that depends jointly on the temporal pattern at each node and the spatial dependencies through the road network. The methods developed for traffic forecasting transfer naturally to commodity logistics in principle, and the framework developed in this thesis is structured to make that transfer concrete. Annex A develops the methodological background in more detail for readers who want to follow the GNN-readiness argument in Chapter 10.

2.3 Energy statistics conventions and the practitioner workflow

The third stream of relevant work is less academic but more directly motivating: the conventions by which energy statistics are published by national and international agencies, and the practitioner workflow that consumes them.

The U.S. Energy Information Administration publishes a wide collection of monthly series covering U.S. crude oil production, refining, stocks, trade, and movements between PADDs. The series are organised by data type (production, stocks, supply, movements) and by geographic resolution (national, PADD-level, refining district, occasionally facility-level). The same physical quantity often appears in multiple series at different resolutions: USA-level production is reported separately from PADD-level production, which is reported separately from named-basin production within each PADD. The series do not always reconcile exactly across resolutions, and the differences typically reflect reporting timing, custody-transfer mechanics, or coverage choices rather than measurement error.

The International Energy Agency, in its monthly Oil Market Report and its annual World Energy Balances, follows a similar pattern at the international level. The IEA explicitly retains a published balancing item in its supply-demand tables, defined as the residual between observed stock changes and the sum of observed flows. The IEA convention treats this residual as information rather than noise: the size and sign of the balancing item carry signal about where the reporting system is failing to close, and the IEA does not silently absorb it into adjacent quantities.

The practitioner workflow that consumes these series is well documented in industry reports and is the immediate motivating context for this thesis. Analysts at trading firms, consultancies, and integrated oil companies maintain spreadsheets or pandas notebooks that load relevant

EIA and IEA series, attempt to balance them against each other across resolutions, and produce derived quantities — implied flows, regional balances, term-structure implications. The workflow is functional but suffers from three known limitations.

First, **reconciliation is implicit and inconsistent across analysts.** Each workbook makes its own hardcoded allocation rules and absorbs the residual into a custom "adjustment" term. Two analysts working on the same underlying data produce different derived quantities because their reconciliation choices differ.

Second, **provenance is lost.** A derived quantity in a typical workbook depends on a chain of allocations and adjustments, and tracing any specific value back to its inputs requires reverse-engineering the workbook. This makes the workflow difficult to audit and difficult to extend.

Third, **multi-resolution mixing is hand-coded.** When a finer-resolution series becomes available — for example, EIA's PADD-level stock decomposition into commercial and SPR portions — incorporating it requires modifying the workbook's allocation logic, often introducing inconsistencies with the legacy resolution that was previously authoritative.

The thesis is, in part, an attempt to systematise this workflow into a structured representation. The twelve principles developed in Chapter 4 are direct responses to each of the three limitations above: schema-level single attribution addresses the first, the resolver's audit trail addresses the second, and the persistent-graph-and-scenario-state model addresses the third.

2.4 Synthesis and the research gap

The three streams reviewed above converge on a clear gap. The petroleum supply chain optimisation literature provides mathematically rigorous models at strategic and tactical levels but operates on abstract network representations that elide the operational detail and multi-resolution structure of the real data. The graph data models literature provides increasingly powerful methods for relational and temporal learning but assumes a clean upstream representation that the commodity context does not naturally provide. The energy statistics conventions describe the data as it is actually published and consumed in practice, but the practitioner workflow that intermediates between the conventions and the downstream consumers is ad-hoc, non-auditable, and difficult to extend.

The gap, stated precisely, is the absence of an upstream representation that:

(a) admits the multi-resolution structure of public energy data without requiring downstream consumers to reconcile it; (b) enforces single-attribution and consistency at the schema level rather than through post-hoc reconciliation; (c) preserves the provenance of derived quantities so that any value in the representation can be traced to its sources; (d) accommodates both classical optimisation consumers (linear programmes) and modern machine learning consumers (graph neural networks) without modification; (e) and is implemented as a working artefact, not only described abstractly.

The thesis fills this gap with the asset-centric temporal graph schema described in Chapter 4, the implementation described in Chapter 5, the resolver described in Chapter 6, the consistency guarantees demonstrated in Chapter 7, and the linear programme consumer demonstrated in Chapter 8. The treatment of graph neural network consumers is deferred to future work and characterised in Chapter 10, where the argument for the framework's readiness is developed.

3. Domain Background: The U.S. Crude Oil Network

This chapter sets out the physical and informational structure of the U.S. crude oil network at the level of detail necessary to make the design choices in subsequent chapters intelligible. The intent is not a comprehensive treatment of the petroleum industry, which would extend well beyond the scope of the thesis, but a focused description of the assets, flows, and reporting conventions that the framework's representation must accommodate. Readers familiar with the industry can skim this chapter; readers approaching the problem from a methods background will find the design choices in Chapter 4 more legible after reading it.

3.1 The physical layer

The crude oil system at the national scale of the United States comprises four physical layers, conventionally described as upstream production, midstream gathering and transportation, downstream refining, and the export and storage infrastructure that interfaces with the international market.

The **upstream layer** consists of the assets that bring crude oil out of the ground. These divide into three structurally distinct categories. Onshore conventional wells, the historical core of U.S. production, are individual oil wells drilled into permeable reservoirs and producing crude through natural pressure or artificial lift. Offshore platforms, concentrated in the federal waters of the Gulf of America (formerly Gulf of Mexico), are large fixed or floating structures producing from sub-sea wells. Shale and tight oil operations, which transformed U.S. production from 2010 onwards, use horizontal drilling and hydraulic fracturing to recover crude from low-permeability formations; their geographic concentration in the Permian (Texas and New Mexico), the Bakken (North Dakota and Montana), and the Eagle Ford (Texas) has shifted the centre of U.S. production decisively to these basins.

The **midstream layer** moves crude from where it is produced to where it is processed or stored. Gathering systems are basin-scale collection networks that pool production from many individual wells into custody-grade volumes suitable for long-distance transport. Trunk pipelines, the workhorses of inland transport, move crude across hundreds to thousands of miles between gathering networks, storage hubs, and refining centres. Tanker shipping (very large crude carriers, Aframax, Suezmax classes) handles long-distance and intercontinental movements, particularly into and out of the Gulf Coast export complex. Rail and truck transport, while smaller in aggregate volume, plays a role in connecting locations that pipelines do not reach (notably during the early Bakken boom before adequate pipeline capacity was built).

The **storage layer** sits between midstream and downstream and absorbs short-term imbalances. Commercial tank farms at hub locations (Cushing in Oklahoma, Patoka in Illinois, the Houston complex, St. James in Louisiana, the LOOP offshore terminal) provide the working storage that refineries and traders draw from. The Strategic Petroleum Reserve, four

underground salt-cavern facilities along the Gulf Coast, provides emergency reserve storage that the U.S. government draws on in response to supply disruptions.

The **downstream layer** transforms crude into refined products. U.S. refineries number in the low hundreds and concentrate heavily in PADD 3 (Gulf Coast), with smaller clusters in PADD 2 (Midwest), PADD 1 (East Coast), PADD 5 (West Coast), and PADD 4 (Rocky Mountains). Refineries vary in complexity (measured by the Nelson Complexity Index), capacity (from under 50 kbd at smaller facilities to over 600 kbd at the largest Gulf Coast plants), and crude slate (the mix of crude grades they are configured to process). The downstream layer is the boundary at which crude oil leaves the modelled system in the form of refined products.

The figure below shows the physical structure schematically, with arrows indicating the direction of crude flow through the chain.

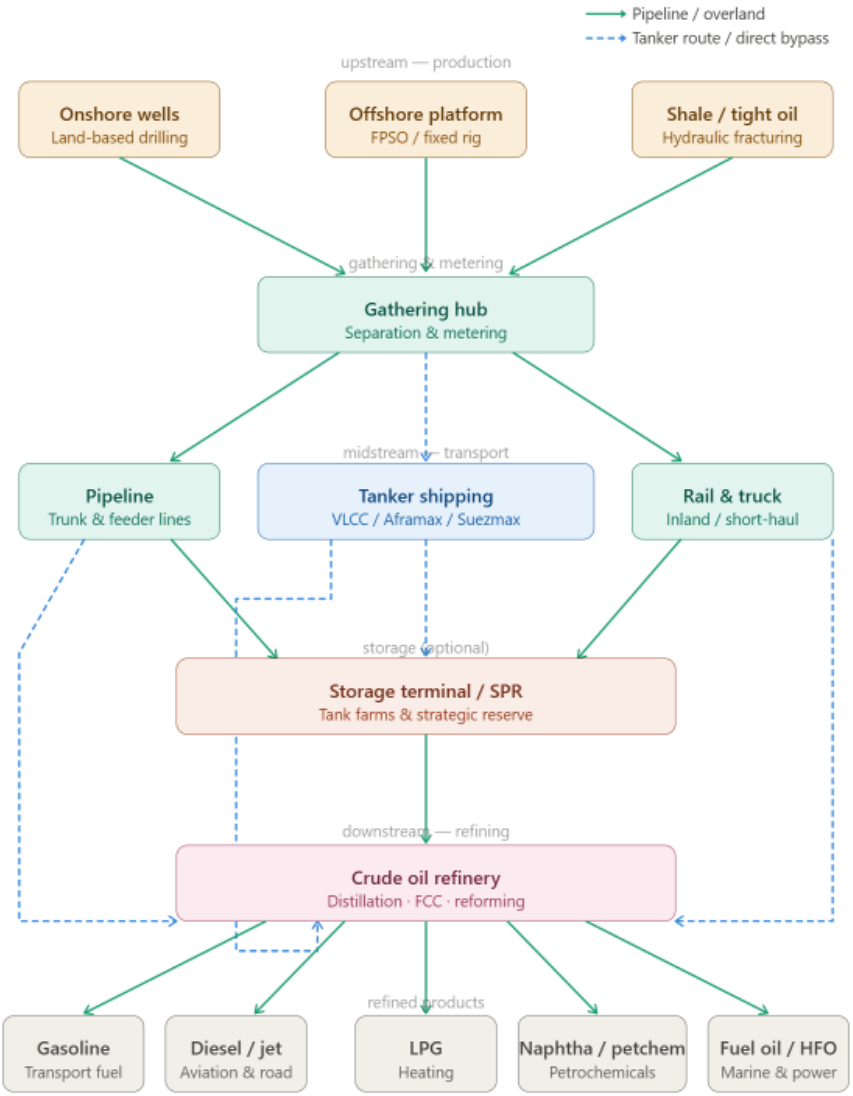


Figure 3.1. The crude oil physical supply chain: upstream production (onshore wells, offshore platforms, shale and tight oil), through midstream gathering and transport (pipelines, tankers, rail and truck), through storage (terminals and SPR), into downstream refining and refined products.

3.2 The PADD structure

The U.S. Energy Information Administration organises U.S. petroleum statistics into five Petroleum Administration for Defense Districts, abbreviated PADDs. The PADDs were created during the Second World War for fuel rationing purposes and have remained the dominant geographic unit for U.S. petroleum reporting ever since. They group the states as follows:

| PADD | Name | States |

|—|—|—|

| PADD 1 | East Coast | Connecticut, Delaware, District of Columbia, Florida, Georgia, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, North Carolina, Pennsylvania, Rhode Island, South Carolina, Vermont, Virginia, West Virginia |

| PADD 2 | Midwest | Illinois, Indiana, Iowa, Kansas, Kentucky, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, Oklahoma, South Dakota, Tennessee, Wisconsin |

| PADD 3 | Gulf Coast | Alabama, Arkansas, Louisiana, Mississippi, New Mexico, Texas |

| PADD 4 | Rocky Mountain | Colorado, Idaho, Montana, Utah, Wyoming |

| PADD 5 | West Coast | Alaska, Arizona, California, Hawaii, Nevada, Oregon, Washington |

Within PADDs, the EIA further subdivides refineries into refining districts (PADD 3A, 3B, 3C; PADD 2A, 2B, 2C; and so on), which group facilities for reporting purposes without corresponding to physical entities. This nested structure is consequential for the framework's design because it imposes a multi-resolution structure on the published data: production might be reported at state-basin level (Permian-TX vs Permian-NM), consumption at refining-district level (PADD 3B), stocks at PADD level, and inter-PADD flows at the PADD-pair level (Gulf Coast to Midwest). Each level is authoritative for some subset of variables, and any representation that wants to consume the data must accommodate this without imposing reconciliation residuals at the boundaries.

3.3 The production geography

U.S. crude oil production in the modelled period (2015 onwards) is dominated by four producing regions, each with distinct geological and operational characteristics:

Permian Basin. Spanning west Texas and southeastern New Mexico, the Permian is the largest U.S. producing basin and accounts for roughly 40 per cent of total U.S. crude production in recent years. Its output exceeded 6 mbd in 2024. The Permian is a shale and tight oil play; its production grew from approximately 1 mbd in 2010 to over 6 mbd in 2024 through the application of horizontal drilling and hydraulic fracturing. The basin is bisected by the Texas-New Mexico state line, and EIA reports the two halves separately (PADD 3 Texas production and PADD 3 New Mexico production), making the basin a natural example of the multi-resolution issue.

Bakken. Spanning western North Dakota and northeastern Montana, the Bakken was the first major shale play to industrialise (peak production around 2014–2015) and remains the second-largest U.S. light tight oil basin. Production peaked at approximately 1.5 mbd before declining; current output is around 1.2 mbd. The Bakken straddles a state line (North Dakota vs Montana), and again EIA reports the two halves separately.

Eagle Ford. Located entirely within Texas, the Eagle Ford shale produces both crude oil and condensate. Its single-state location simplifies the reporting picture: Eagle Ford production is the entirety of certain EIA Texas series.

Gulf of America offshore. Federal waters in the Gulf produce approximately 1.8 mbd in the modelled period, dominated by deepwater platforms operated by major integrated oil companies. EIA reports Gulf production as a single series rather than disaggregating by platform or field.

Beyond these four, smaller volumes are produced in Alaska (the trans-Alaska pipeline from the North Slope), in the conventional Oklahoma fields, in California, Wyoming, and elsewhere across the country. EIA publishes most of these at state level, with smaller un-named producing volumes within each PADD captured by a PADD-level total that exceeds the sum of named contributors. This residual — the difference between the PADD total and the sum of named basins — is the motivating example for the residual-node design choice described in Chapter 5.

3.4 The transport network

Trunk pipelines connect the producing basins to the refining centres and to the export complex. The network is dense enough that no production region is more than a few hundred miles from a major refining cluster, but the pipeline-by-pipeline structure matters for understanding where data is observed and where it is not.

Permian outbound. The Permian's six-million-barrel daily output exits the basin through three origin terminals: Midland (the largest, in the heart of the basin), Wink (to the southwest), and Crane (to the southeast). From each origin terminal, multiple trunk pipelines run to the Gulf Coast: Gray Oak (Wink and Crane to Corpus Christi), Cactus II (Midland to Corpus Christi), EPIC (Midland to Corpus Christi), Wink-to-Webster (Wink and Midland to Houston), and others. EIA reports Permian production at basin level and Gulf Coast receipts at terminal level, but the per-pipeline split between Permian and the receiving terminals is not reported in public data. This is the canonical example of the latent-flow problem described in Chapter 4.

Bakken outbound. The Bakken connects to PADD 2 (Midwest) and onward to other PADDs through a network of pipelines (Dakota Access being the dominant trunk) and rail terminals. The per-pipeline split out of Bakken gathering is again unobserved in public data.

The Cushing complex. Cushing, Oklahoma is the largest crude oil storage hub in the U.S. and the delivery point for the WTI futures contract. Pipelines converge on Cushing from all

directions: north from Texas (Seaway north-bound), south to the Gulf (Seaway south-bound, MarketLink, Keystone XL, Cushing Marketlink), east to Patoka (Diamond, Mid-Valley), and into Cushing from the north (Pony Express from the Bakken via Wyoming). The Cushing hub itself comprises multiple operator sub-terminals (Enterprise, Plains, Magellan, and others), each reporting separately at quarterly intervals but aggregated to hub level monthly.

Inter-PADD flows. Crude movements between PADDs are reported by EIA at the PADD-pair level (PADD 2 to PADD 3, PADD 3 to PADD 1, and so on), without disaggregating which specific pipelines carry which volumes. The framework's inter-PADD aggregate nodes are direct responses to this reporting choice: each aggregate gives the partition tree a same-type child to point at, with the per-pipeline decomposition latent for future LP allocation.

The Gulf Coast export complex. Approximately 4 mbd of U.S. crude leaves the country through Gulf Coast export terminals as of 2024, predominantly via the Houston complex (Enterprise, Magellan, Moda Midstream, and other operators), the Corpus Christi cluster (Ingleside being the largest single terminal, primarily handling Permian crude), Nederland in southeast Texas (Sunoco), LOOP in offshore Louisiana (the only U.S. terminal capable of fully loading VLCC class tankers), and St. James in Louisiana. Each terminal has its own operator-reported throughput, but the per-destination breakdown (where the exports go after leaving the terminal) is reported at the country-destination level rather than terminal-destination, again creating a multi-resolution issue.

3.5 Refining and consumption

U.S. refining capacity is approximately 18 mbd, with consumption (crude inputs to refineries) typically running at 15–17 mbd. The geographic concentration is heavy: PADD 3 (the Gulf Coast) contains roughly half of all U.S. refining capacity, and within PADD 3 a small number of integrated coastal facilities (Motiva Port Arthur, ExxonMobil Beaumont, Marathon Galveston Bay, the Valero plants) each exceed 400 kbd of capacity. PADD 2 (the Midwest) has the second-largest concentration, including refineries supplied predominantly from Canadian heavy crude (BP Whiting, Marathon Detroit) and others supplied from a mix of domestic and Canadian crude.

Refineries are characterised by their throughput capacity (the maximum crude they can process), their configuration (the units they contain: crude distillation, coking, hydrocracking, fluid catalytic cracking, alkylation, reforming), their complexity (a single Nelson index summarising the configuration), and their crude slate (the mix of crude grades they are designed to process efficiently). Slate flexibility is finite and operationally significant: a refinery designed for heavy sour Canadian crude cannot simply switch to light sweet Permian crude without significant operational implications.

For the present framework, refineries are nodes with three primary variables: consumption (crude throughput, observed at the refining district or sometimes facility level), inventory (operating stocks, observed at PADD level), and a balancing item that absorbs process losses

and reporting residuals. The detailed product slates and the propagation of crude grade through the refining process are out of scope for the initial implementation, which treats crude as a single commodity.

3.6 What the framework must accommodate

The description above motivates several specific requirements that the framework's representation must satisfy:

Multi-resolution data without reconciliation residuals. Production observed at basin level, stocks at PADD level, consumption at refining-district level, and inter-PADD flows at PADD-pair level all need to coexist in a single representation. The framework must not require any of these to be "down-disaggregated" to a uniform finest-granularity before being usable.

Junction fan-outs with unobserved per-edge splits. The Permian outflow problem (three terminals, multiple pipelines from each, downstream destinations observed only at aggregate level) is structural, not exceptional. Bakken gathering, the Cushing hub, and the Gulf Coast export complex all exhibit the same pattern at different scales. The framework must handle these natively rather than treating them as edge cases.

A balancing item that absorbs systematic reporting residuals. The IEA convention of a published balancing item exists for a reason: the data does not close exactly, and the residual carries information about where the reporting system is failing to close. The framework must retain B as a first-class variable rather than absorbing it into adjacent flows.

Slow structural evolution. The pipeline network changes deliberately over time (Capline reversed in 2022, the Bakken cross-state connector was added in 2024). These are real operational events, not data artefacts, and the framework must accommodate them without requiring a wholesale graph rebuild.

A boundary that separates the modelled system from the rest of the world. Foreign supply comes in through Canadian and other import flows; U.S. crude leaves through export terminals to destinations the framework does not track. The boundary needs to be explicit, with mass balance closing on the U.S. side without requiring the framework to model the foreign side.

These requirements are not optional; they reflect the structure of the data as it is actually published and the structure of the system as it actually operates. The next chapter develops the twelve principles that respond to these requirements directly.

4. Design Principles

This chapter sets out the twelve architectural principles that govern the asset-centric oil-network schema. Each principle is presented in the same form: the rule itself, the alternative that was genuinely on the table, the rationale that decided between them, and a concrete example drawn from the U.S. crude case study. The principles are not aspirational guidelines but ratified commitments, each enforced either by the database schema, by the resolver, or by review during construction.

The principles together encode the consistency guarantees that this thesis claims as its contribution: an asset-centric representation that admits multi-resolution data, preserves provenance through the resolution chain, and detects double-counting at insert time rather than after the fact. The presentation order follows the natural dependency among the principles, beginning with the representational commitment that the rest build upon.

4.1 Asset-centric representation

The first commitment is that every physical asset in the network is a node carrying its own inventory and mass balance. Pipelines, vessels, terminals, refineries, gathering networks, and storage hubs are first-class nodes, not properties of the edges between them. The alternative under serious consideration was a location-centric representation, in which nodes denote geographic places (basins, refining centres, ports) and the assets sit on the edges that join them. Both representations are common in the supply-chain optimisation literature; the choice has consequences that are easy to under-appreciate until edge cases force them into view.

The location-centric view treats a pipeline as an attribute of an edge: a connector with length, capacity, transit time, and operational status. Under this view, the crude oil physically inside the pipeline at any given moment, the line-fill, is hidden state. It must be reconstructed implicitly from the edge's transit time and the recent inflow rates. For most edges in most networks this is a perfectly serviceable abstraction. For a crude oil network it is not. A 36-inch trunk line running 1,200 miles at 600 kbd holds something on the order of five days of throughput in line-fill, often comparable in volume to a regional refinery's weekly throughput. When the line throughput changes, when the line shuts down for maintenance, when a hurricane forces a temporary suspension, the line-fill moves. That movement appears in regional inventory statistics. Treating it as edge attribute hides the movement; treating the pipeline as a node makes it visible.

The same argument applies, more strongly, to vessels. A VLCC carrying two million barrels from Rotterdam to Houston is a floating storage facility for the three weeks it takes to cross the Atlantic. Its cargo is real inventory in transit, and during periods of contango the practice of slow-steaming or holding vessels offshore as effective storage becomes economically rational. A representation that hides this volume in an edge attribute mishandles a structurally important part of the system.

The asset-centric commitment is therefore: each pipeline, each vessel, each terminal is a node with its own stock variable, its own mass balance, and its own potential observation series. Edges become what they physically are — directed flows from one asset to another, carrying volume but not holding it.

Example. When the Capline pipeline reversed direction in 2022, switching from south-bound to north-bound service, the asset-centric representation handles the change naturally. The pipeline remains a node. Its line-fill remains its own stock variable. The two directed edges that connect it to the Patoka hub at one end and the St. James terminal at the other remain in the topology; one direction goes to zero flow, the other becomes active. The location-centric alternative would have required a structural rewrite of the edge: changing its direction, possibly renaming it, and reconstructing how its line-fill is attributed across the change. Asset-centric makes the reversal a data event, not a schema event.

A second example, from the live implementation, demonstrates the same point quantitatively. EIA's stock identity for PADD 5 — commercial stocks plus the Strategic Petroleum Reserve plus in-transit volumes — consistently overshoots the sum of named commercial hubs and refinery stocks by approximately six million barrels. The gap is the inventory of crude in transit aboard Jones Act tankers carrying Gulf Coast crude to West Coast refineries, which EIA allocates to PADD 5 as destination-based ending stocks. In an asset-centric model this volume has a natural home: the inter-PADD pipeline corridor node from PADD 3 to PADD 5 already exists with TS-bound flow variables, and the residual is placed as its line-fill inventory, closing PADD 5's stock partition exactly. A location-centric model, where the corridor is an edge attribute rather than a node, would have nowhere to put those barrels.

4.2 Stable topology via zero-flow edges

The second commitment is that the adjacency matrix of the graph is fixed across time. Every edge that could ever carry flow under the modelled topology is defined once, at construction time, and remains present in the graph at every timestep regardless of whether it is currently active. Inactive edges carry zero flow, not absent flow.

The alternative is a time-varying topology, in which edges appear and disappear as the underlying physical or contractual relationships change. This is the natural representation if one thinks of the graph as a literal record of which flows are happening at a given moment. It is also the representation that many dynamic graph methods in the machine-learning literature assume. The reason this thesis rejects it is twofold. First, a stable adjacency matrix is a precondition for the standard graph neural network architectures (GCN, GAT, GraphSAGE, and the temporal extensions DCRNN, STGCN, TGN) that are the natural downstream forecasting consumers of this representation. Architectures that handle graph rewriting at every timestep exist but are markedly more complex and less mature. Second, a stable topology cleanly separates structural change (rare, deliberate, requiring schema review) from

operational change (frequent, automatic, reflected in flow values). The two kinds of change have different causes and demand different review processes.

The zero-flow convention is therefore an act of design hygiene rather than a physical statement. A pipeline that is shut for maintenance still exists as a node; the edges into and out of it still exist; the flows on those edges are zero for the duration of the outage. A reversible pipeline carries two directed edges in opposite directions; at any moment, one or both flows may be zero. The graph topology never changes; only the values on the edges change.

Example. The Capline reversal mentioned above illustrates the rule. Both directed edges, north-bound and south-bound, exist throughout the modelled period. From 1944 (when Capline opened) through 2021 the south-bound flow was active and the north-bound flow was zero. From 2022 onwards the relationship flipped: the north-bound flow lit up, the south-bound went to zero. No structural rewrite of the graph was required, and no GNN architecture that depends on a stable adjacency matrix needed special handling.

4.3 Universal mass balance with balancing item

The third commitment is that every node, regardless of type, satisfies the same mass balance equation:

$$\Delta S(i, g, t) = P(i, g, t) + F\_in(i, g, t) - C(i, g, t) - F\_out(i, g, t) + B(i, g, t)$$

where for node i, commodity g, and time t, ΔS denotes the change in stock, P is production, C is consumption, F_in is total inflow from connected nodes, F_out is total outflow to connected nodes, and B is the balancing item. The balancing item is a first-class node variable: a recorded value that absorbs systematic discrepancies between the observed stock change and the sum of the observed flows. It is retained as part of the node's data, not discarded as noise.

The natural alternative is to derive B implicitly as a residual at query time, or to treat any non-zero residual as a data quality problem to be cleaned up before the data is loaded. Both alternatives have been tried; both fail in the crude oil context for the same reason. Reporting frictions exist at every reporting boundary — basin-to-pipeline custody transfers, refinery process losses, vessel cargo gains and losses, measurement timing offsets — and discarding the residual silently propagates those discrepancies into derived quantities downstream. The IEA, in its World Energy Balances, retains a published balancing item for exactly this reason. The convention is not a confession of measurement failure; it is a deliberate decision to keep the residual visible.

In the present framework the balancing item carries additional weight. Its size and sign are operationally informative. A persistent positive B at the USA aggregate level indicates that observed stock changes exceed what the reported flows would imply, which can be a signature of unreported imports, under-reported exports, or measurement timing offsets. A sharp transient spike in B around a known event — a hurricane disrupting Gulf Coast

production, a pipeline outage, a major SPR release — measures the magnitude of the data gap that the event opened up. Chapter 7 develops this into a consistency claim of its own, comparing observed B against the closure-derived B as a way of localising where the framework's representation diverges from the public data.

Example. Refineries, considered individually, have small but non-zero B values reflecting process losses (the well-known refinery shrinkage of approximately 1 to 2 per cent by volume). Pipelines have small B values reflecting line loss and measurement noise. Production nodes have small B values reflecting reservoir condensate and flaring. At the USA aggregate level, B is typically a few tens of kbd in absolute value, rising to several hundred kbd around named disruption events. Each of these B values is a retained data point, available for inspection and for downstream consumers; none of them is silently absorbed into the surrounding flow variables.

4.4 Formula-implies-relation

The fourth commitment is that edges, partition links, and node status are all derived views of the variables collection. The single source of truth is the variables table; every graph view is a database `CREATE VIEW` over it. An edge between two nodes exists if and only if there is a variable on one node whose formula references a variable on the other. There is no separate edges table whose rows must be maintained in lockstep with the variable definitions.

The alternative — and it was the original design in early implementation passes — is to maintain edges as a primary declared structure: a separate table whose rows say "edge from node A to node B exists, with these attributes". The trouble with this alternative is that it permits structural inconsistency between the edge table and the variable definitions. A declared edge with no corresponding variable formula is useless; a variable formula referencing a node with no declared edge is mathematically defined but graphically invisible. Either failure mode is recoverable, but only by carrying two parallel structures and reconciling them. Formula-implies-relation eliminates the second structure entirely: if the variables collection is consistent, every graph view is consistent by construction.

The same principle extends to the resolution hierarchy (the same-type aggregation tree, see Section 4.7) and to the per-scenario node status (see Section 4.5). In each case the structural artefact is recoverable from the variable definitions and is therefore not stored a second time.

Example. Four views of the graph are derived from the single variables table: `v_flow_edges` lists every directed flow edge in the network, derived from the relational variables that carry a `related_node_id`; `v_aggregation_edges` lists every cross-node formula reference, regardless of variable type; `v_partition_tree` filters the aggregation edges to the same-type, same-commodity subset that defines the resolution hierarchy; and `v_node_status` (see Section 4.5) classifies each node under the active scenario based on the assignment rows attached to its variables. None of these views requires its own primary table; all are refreshed whenever the variables collection or the scenario assignments change.

4.5 Persistent asset graph and scenario state

The fifth commitment separates the model into two layers. The persistent asset graph is the fixed, superset physical topology — every node and every feasible edge defined once and never altered for data reasons. The scenario state is derived at load time from the persistent asset graph plus the per-scenario `variable_assignments` rows, which specify, for each variable under that scenario, whether it is bound to a time series or to a formula.

The alternative is a single-layer model in which any change in data resolution (a new EIA series at finer granularity, the introduction of facility-level telemetry, per-grade decomposition) rewrites the graph. Single-layer was the original design and proved unsustainable. Every data improvement forced a structural revision; every structural revision broke downstream consumers; and the lineage of any given derived quantity became impossible to trace across the changes.

The two-layer model puts the volatility in `variable_assignments`. The persistent asset graph stays stable; data improvements re-point variables from coarser to finer time series within the existing node set. If the existing node set is insufficient, the persistent asset graph is extended (a new node is added, an existing node is split) by an explicit migration, and the migration is recorded as a versioned event. The asset graph evolves slowly and deliberately; the scenario state evolves with the data.

The status of each node under a given scenario is itself derived from the assignment rows attached to that node. A view `v_node_status` classifies each node as **authoritative** if at least one of its variables is time-series-bound, **derived** if no variable is time-series-bound but at least one resolves through a non-trivial formula, and **collapsed** if every variable on the node is either structurally zero or latent. The three-way classification is computed, not declared; if the variable definitions are consistent, the classification is consistent.

Example. When per-refinery operating stocks become available for the first time, the framework does not require an asset graph revision. The relevant refinery nodes already exist; what changes is that their inventory variables are re-pointed from PADD-level aggregate time series to facility-level time series in `variable_assignments`. The refineries' status under the scenario flips from collapsed to authoritative automatically through `v_node_status`. No code outside the assignment table changes.

4.6 Physical layer and observational aggregation layer

The sixth commitment is that the physical layer of the graph (real named assets — fields, pipelines, vessels, terminals, refineries) is structurally distinct from the observational aggregation layer (PADDs, refining districts, basin roll-ups, national totals). The aggregation layer carries no edges in the physical graph; it materialises rollups on demand through same-type aggregation formulas.

The alternative is to mix administrative and physical entities in a single layer. PADD 2 would then be a node like Cushing or the Bakken basin, with flow edges directly to and from it. The trouble with this alternative is that PADD 2 is not a physical entity. It has no operator, no equipment, no custody transfer mechanics; it is a reporting boundary used by the U.S. Energy Information Administration to group physical assets within the central United States. Treating it as a physical node invites confusion: does the PADD have a pipeline to its neighbour, or is that pipeline aggregated from the underlying physical pipelines? The framework picks the latter unambiguously. PADD-level flows are alias-derived from the corresponding inter-PADD aggregate nodes' time-series-bound outflows; PADD-level production is sum-derived from the underlying physical basins; PADD-level consumption is sum-derived from the underlying refineries.

A subtler benefit of the separation is that observational aggregates can be added, modified, or removed without affecting the physical layer. EIA refining-district roll-ups (PADD 3A, 3B, 3C, …) exist because EIA publishes district-level series; they are aggregation views over refineries, not co-owned operating entities. If EIA changed its reporting structure tomorrow, the framework would add or retire aggregation nodes accordingly, without touching a single physical asset.

Example. The node `padd2_view` aggregates the Bakken-ND basin, Oklahoma, the padd2_other residual, the `padd2_canadian_imports_agg` import aggregate, the inter-PADD corridor aggregates with PADD-2 endpoints, and twelve named refineries. It has no flow edges of its own. Its outflow to PADD 3 is alias-derived from the time-series-bound outflow on the `inter_padd_2_to_3_agg` node, which is itself a physical aggregate over the named inter-PADD pipelines (Capline, Diamond, ExxonMobil Pegasus, and others).

4.7 Geography metadata distinct from resolution hierarchy

The seventh commitment distinguishes geography metadata — properties describing where a node sits, such as latitude, longitude, county, state, PADD, country — from the resolution hierarchy, which is a structural relationship between nodes representing the same physical system at different granularities. The Permian basin contains the Texas portion of the Permian, which contains Permian-Midland County, which contains individual leases. This is a resolution hierarchy; it is enforced structurally. Two refineries that both sit within PADD 3 share a geography label; they are not part of the same physical asset, and the geography label does not make them so.

The distinction is important because labels cannot prevent double-counting on their own. Assigning observed data to multiple levels of the same hierarchy for the same variable would double-count; the framework rejects this at the observed/derived invariant level (Section 4.8). Geography metadata is for filtering and rollup queries; hierarchy is for resolution. Conflating the two — treating "is in PADD 3" as if it implied a parent-child relationship in the data — is the failure mode that this principle is designed to prevent.

Example. Two named refineries can share `padd = 'PADD3'` in the geography metadata without being part of the same physical asset (and they typically are not — Motiva Port Arthur and ExxonMobil Beaumont are independent operating entities that happen to share the PADD label). Conversely, two nodes can sit at `resolution_hierarchy.parent = 'permian'` (the basin) without being co-located in geography (Permian-TX is in Texas, Permian-NM is in New Mexico). The two metadata systems are independent.

4.8 The observed/derived invariant

The eighth commitment is the single most important consistency guarantee in the framework. For each (node, variable type, commodity, timestamp), exactly one of the following holds: the variable is **observed**, meaning it is bound to a time series and the time series value is the authoritative value at that resolution; or the variable is **derived**, meaning it is bound to a formula and its value is computed from other variables. A variable cannot be both. A variable cannot be neither.

The alternative is post-hoc validation: allow any combination of bindings, and run a reconciliation pass after the fact to detect double-counted attributions. Post-hoc validation has the same defects in every domain it appears in. It detects the problem only after the data has propagated downstream; it requires an expensive scan over the entire data set; it produces warnings rather than rejections; and it admits the temptation to silence the warnings rather than fix the underlying assignment. Schema-level enforcement, by contrast, makes the inconsistent state unrepresentable. The relevant CHECK constraint is:

```
CHECK (num_nonnulls(timeseries_id, formula) = 1)
```

It cannot be violated by an INSERT. Any attempt to write an assignment row that is both time-series-bound and formula-bound, or neither, is rejected by the database before the data can propagate anywhere.

A complementary audit at the time-series attribution level enforces single-attribution on the other side: each EIA time series is bound to exactly one variable under a given scenario. Together, the two invariants give the framework its central consistency property: every value in the resolved table is traceable to exactly one source, and no source contributes to more than one variable.

The framework does allow auxiliary observed series on the same node — for example, EIA publishes both `MCRSFP` (PADD-level stocks excluding SPR) and `MCRSSP` (SPR-level stocks) — but these are marked as **observed auxiliary** rather than **observed authoritative**, and they do not participate in mass balance at their own level. They serve as cross-checks against the authoritative value.

Example. When EIA began publishing the decomposition of PADD-level stocks into Strategic Petroleum Reserve and commercial portions (MCRSFP plus MCRSSP), the natural

temptation was to bind each component at a separate sub-node level. But the SPR is already a structural node in the asset graph, with its own four physical sites and its own inventory variable, and MCRSFP at PADD level would have double-counted against the existing per-site SPR values. The framework forced the choice: MCRSFP is registered as an observed auxiliary series on the PADD view, not as an authoritative binding on a separate sub-node. The pre-existing SPR sub-nodes remain the authoritative source for their own inventory.

4.9 Latent allocation at junctions

The ninth commitment governs the treatment of flows that are real but unobserved. At many junction nodes in a real network, flows split or merge in ways that public data does not directly measure. The Permian basin produces approximately 4,300 kbd of crude; that volume fans out across three origin terminals (Midland, Wink, Crane), each routing onward through multiple pipelines (Gray Oak, Cactus II, EPIC, Wink-to-Webster, and others), each of which terminates at one of several Gulf Coast destinations. The total flowing out of Permian is observed; the total arriving at each destination is observed; the per-pipeline split is not.

The framework treats the unobserved per-edge flows as **latent**: declared variables with a known structural role and an unknown value. The mass balance constraint at the junction (total outflow equals total inflow plus production minus consumption) still applies; it becomes a constraint that the latent variables must collectively satisfy. The alternative — assigning per-pipeline values via heuristic rules such as proportional capacity weighting — has the seductive virtue of producing complete data. The alternative is rejected because the heuristic values are not data; they are guesses that downstream consumers cannot distinguish from real measurements.

Latent variables are first-class entities in the resolved table. They appear with `value = NULL` and `source = 'latent'`, distinguishing them from `unresolved` rows (which represent a resolver failure and should never occur in a healthy run). Downstream consumers can therefore distinguish unknown-by-design from unknown-by-failure. The natural consumer for the latent variables is a linear programming model, treated in Chapter 8: the latents become decision variables, the mass balance constraints become equality rows in the LP, and the resulting system has a well-posed solution under standard optimisation objectives.

Pass-through node types (gathering networks, pipelines, import and export terminals, foreign aggregates) have P = C = ΔS = B = 0 by physical construction, encoded in `node_type_default_formulas`. This makes the mass balance at every junction reduce to ΣF_in = ΣF_out, which is the natural form for the LP to consume.

Example. The node `permian_tx` has an observed production of approximately 4,300 kbd in a recent month. It connects to three origin terminals (`midland_terminal`, `wink_terminal`, `crane_terminal`) whose individual receipts from Permian are not publicly reported. The three outflow variables from `permian_tx` are declared latent. The constraint that they sum to the observed production is enforced by the mass balance at `permian_tx`. The gathering nodes

for the Bakken (one node, seven downstream PADD-2 destinations, individual splits unobserved) and the Cushing hub (three operator sub-terminals, sub-decomposition observed by EIA but not at all desired granularities) follow the same pattern at different scales.

A second example concerns the Bakken cross-state midstream. The North Dakota and Montana gathering systems are operated commercially as a single pool by Hess, Crestwood, and Energy Transfer, with crude moving between the two states in both directions depending on takeaway capacity. Public data does not publish the per-direction flow. The framework models this as a bidirectional pipeline node pipe_bakken_xstate carrying four flow variables — two inflows and two outflows, one per direction at each gathering — all latent. The mass-balance constraint at the connector is preserved by pipeline-type defaults ($P = C = \Delta S = B = 0$, so $\Sigma I = \Sigma O$). The connector is declared as a constituent of the inter-PADD aggregate outflows on both sides, so future per-operator gathering telemetry lands on these variables without any topological change.

4.10 Bidirectional flows as separate directed edges

The tenth commitment is that each direction of a reversible or bidirectional connection between two nodes is modelled as a separate directed edge, each with its own flow variable. The zero-flow convention from Principle 2 handles the inactive direction at any given moment.

The alternative is a single undirected edge with a signed flow variable: positive in one direction, negative in the other. This is the more compact representation in terms of variables defined, and it appears in some optimisation models for transportation networks. The reason this thesis rejects it is operational: a south-bound flow on Capline (Patoka to St. James, delivering crude into Gulf Coast refining) and a north-bound flow on the same pipeline (St. James to Patoka, delivering crude into mid-continent storage) are physically distinct operations with different tariff structures, different congestion patterns, different operator commitments, and different downstream consumers. Conflating them under a single signed variable obscures these differences.

The directed-edge convention also interacts cleanly with the zero-flow convention. At any moment, at most one direction on a reversible pipeline is active. Both directed edges remain in the topology; the active one carries the current flow; the inactive one carries zero. When the pipeline reverses, the values swap. The structure is unchanged.

Example. The Seaway pipeline carries two flow variables: `outflow__crude__pipe_seaway__houston_hub` (south-bound, the dominant direction) and `outflow__crude__pipe_seaway__cushing_hub` (north-bound, briefly active under specific market conditions). The same applies to Capline (reversed in 2022, both directions now in the topology), the LOOP offshore terminal, and the SPR sites (which can receive crude for storage or deliver crude in response to a release). The recently-added cross-state Bakken connector (`pipe_bakken_xstate`) follows the same pattern with four flow variables (two inflows, two outflows, all currently latent).

4.11 The canonical sum rule

The eleventh commitment is more technical than the previous ten and reflects a lesson learned through implementation rather than a high-level design choice. In an earlier version of the resolver, three distinct dispatch labels were used for sum-type derivations: `sum_over_children` (for aggregation parents whose inputs are same-type children), `sum_over_outflows` (for fan-out totals whose inputs are different-type outflows at the same node), and `sum_same_type` (a near-duplicate of `sum_over_children` used in specific contexts). The three labels encoded slightly different semantic roles but produced identical numerical results.

The twelfth implementation pass collapsed all three into a single canonical `sum` rule. The semantic role of any given sum — aggregation, fan-out, residual definition — is recoverable from the structure of its `formula_inputs`, and so the formula text does not need to encode it. Specifically, if all inputs share the same variable type, commodity, and related node as the parent, the sum is an aggregation parent. If all inputs are outflow variables at the same source node but with different target nodes, the sum is a fan-out total. If the inputs mix variable types, the formula is not a sum at all but an arithmetic residual (handled by a separate rule, see Section 4.12).

The lesson generalises beyond sums and is worth stating as a principle in its own right: label proliferation in the resolver is a symptom of carrying structural information in formula text that should be carried in formula inputs. The same lesson drove the consolidation of several other dispatch paths in later implementation passes.

Example. Three different formulas in the starter scenario fire the canonical sum rule. The aggregation `padd1_view.P = Σ(P over all production nodes in PADD 1)` is a same-type roll-up: all inputs are production variables with PADD-1 nodes. The fan-out `midland_terminal.F_out_total = Σ(outflow to each downstream pipeline)` is a different-type set: each input is a directional outflow to a different target. The residual `padd2_other.P = padd2_view.P − bakken_nd.P − oklahoma.P` is not a sum at all; it is an arithmetic combination, and the resolver dispatches it through the arithmetic rule, not the sum rule.

4.12 Latent and unresolved are distinct

The twelfth commitment closes the framework's treatment of missing values. Both `latent` and `unresolved` rows in the resolved table carry `value = NULL`, but they mean different things and the framework treats them differently.

A `latent` row is a deliberate declaration: the variable exists structurally, its value is unknown by design (typically because the per-edge split at a junction is not publicly observed), and downstream consumers should treat it as a free variable constrained by mass balance. The natural consumer is an LP that uses the latent as a decision variable.

An `unresolved` row is a failure: the variable has a formula, but the resolver could not evaluate it. The typical cause is a referenced variable being missing from the assignment set, or a formula pattern that the resolver does not yet recognise. The expected operator response is to fix the assignment (correct the reference) or extend the resolver (add a rule for the new pattern). An unresolved row is never silently treated as latent.

The distinction matters because it tells downstream consumers what to do. A latent row is data; an unresolved row is a bug. In a healthy run of the resolver, the count of unresolved rows is zero, and this fact is the load-bearing invariant on which Chapters 6 and 7 rest.

Example. In the starter scenario at the time of writing, 652 variables resolve to `latent` and 0 resolve to `unresolved`. The 652 latents are concentrated at junction fan-outs: the three Permian outflows, the seven Bakken gathering outflows, the multiple offshore Gulf production routes, and the per-pipeline splits inside each inter-PADD aggregate. The zero unresolved is the guarantee that every variable has either a value or an explicit latent marker; no variable is silently missing.

4.13 Summary

The twelve principles together specify what the framework is and what it is not. The framework is asset-centric, single-source-of-truth, schema-enforced, multi-resolution, and explicit about its unknowns. It is not a forecasting system, not a flow-allocation heuristic, and not an attempt to clean noisy data into completeness. The next chapter develops the schema that gives effect to these principles, and Chapter 6 details the resolver that consumes the schema and produces the per-scenario data tensor on which the LP and any future forecasting consumer will operate.

5. Schema and Graph Construction

The principles set out in Chapter 4 are commitments at the architectural level. This chapter describes the schema and construction process that give effect to those commitments: the two-layer model, the taxonomy of nodes, the structure of variables and edges, the partition tree that drives consistency validation, and the starter graph that instantiates all of this for the U.S. crude oil network from 2015 onwards.

5.1 The two-layer model

The framework separates a persistent layer from a scenario layer. The persistent asset graph is the fixed, superset physical topology — every node, every potential edge, defined once and not altered for data reasons. The scenario state is derived at load time from the persistent asset graph plus the per-scenario rows in `variable_assignments`. Downstream consumers query the scenario state, not the persistent graph directly.

The separation is the practical expression of Principle 5. Data volatility lives in the assignment table; structural volatility, when it occurs, is captured by an explicit migration to the persistent graph. The two kinds of change are reviewed differently, deployed differently, and rolled back differently. A new EIA series at finer granularity is an assignment change. The construction of a new export terminal is a migration. The framework refuses to conflate them.

In practical terms, an early framing of the work included a separate object called a "coverage contract" that recorded which level of resolution was authoritative for each subsystem under a given scenario. As the implementation matured, it became clear that the coverage contract was mechanically identical to the assignment table tagged with a `scenario_id`. The choice of authoritative level for each variable is exactly the per-variable selection of a time series binding or a formula binding. No separate object was needed; the principle was preserved, the structure was simplified.

5.2 The starter asset graph

The starter graph targets U.S. crude oil from January 2015 onwards, the period over which the relevant EIA monthly series are consistently available. It contains 240 assets in total: 197 physical and 43 abstract aggregates, of which 4 are boundary nodes. The assets connect via 409 directed flow edges, and they carry 1,830 variables under the active scenario `starter_us_crude_2015_2025`.

Every node is tagged with a `kind` (physical or abstract), a `node_class` (production, infrastructure, observational), and a `node_subtype` that refines the class (refinery, gathering, pipeline, region_view, and so on). Geographic metadata — latitude, longitude, state, county, PADD — is stored in a separate `oil_network.locations` table to keep labelling distinct from structure, in line with Principle 7.

The graph is built from a seed JSON file (`asset_graph.json`) loaded once and then evolved by incremental migration scripts. Each migration corresponds to a deliberate structural change (a new aggregate node, a node split, a route correction) and is recorded as a versioned event. The final state of the starter graph reflects twelve such migration passes.

5.3 Physical nodes

Physical nodes represent real, named assets on the ground. They carry geographic coordinates, capacity, configuration, and flow edges. Adding a physical node usually corresponds to a real operational change: a new refinery coming online, a new pipeline being commissioned, a new export terminal opening.

The starter graph contains the following physical inventory, grouped by subtype:

| Subtype | Count | Role and representative examples |

|—|—|—|

| state_sub_basin | 5 | Named production regions: bakken_nd, bakken_mt, permian_tx, permian_nm, eagle_ford_tx |

| state_conventional | 7 | Mature state-level production: oklahoma_conventional, california_conventional, wyoming |

| state_residual | 5 | One per PADD: padd1_other through padd5_other, covering un-named tail production |

| offshore_region | 1 | gulf_of_america (encompassing all federal offshore production in the Gulf) |

| gathering | 6 | Basin-level midstream pooling: ans, bakken_nd, bakken_mt, eagle_ford, permian_nm, permian_tx |

| origin_terminal | 6 | First-stage receipt points: midland, wink, crane, johnsons_corner, three_rivers, valdez |

| storage_terminal | 10 | Cushing hub (with three operator sub-terminals), Patoka hub, Houston, and others |

| spr_site | 4 | The four Strategic Petroleum Reserve sites: bayou_choctaw, big_hill, bryan_mound, west_hackberry |

| refinery | 115 | 19 named refineries, 5 PADD residuals, and sub-aggregates totalling 24 active facilities |

| import_terminal | 8 | clearbrook plus four PADD-level import aggregates, two Canadian import aggregates, padd2_xstate |

| export_terminal | 12 | Ingleside, Houston-aggregate plus three sub-terminals, Nederland, LOOP, and five PADD-level export aggregates |

| pipeline | 37 | 20 named U.S.-internal lines, 4 cross-border lines, 12 inter-PADD aggregates, the new Bakken cross-state connector |

| foreign_export_destination | 1 | Boundary sink: where U.S. exports physically leave the modelled system |

| foreign_production_aggregate | 2 | Boundary source: canadian_oil_sands and foreign_supply (non-Canadian) |

A few of these categories warrant comment. The **state residuals** (padd1_other through padd5_other) exist because EIA publishes PADD-level production totals that exceed the sum of the named basins and conventional fields within each PADD. The difference must be attributed to some node; the residual is that node. Each state residual carries an explicit list of the production it stands in for, so that when new physical sub-basins become individually named, the residual shrinks correspondingly and the substitution is traceable.

The **inter-PADD and import/export aggregates** are technically physical nodes — they have `kind = 'physical'`, an asset row, and concrete flow edges — but they aggregate over groups of named pipelines and terminals rather than denoting a single facility. They exist to give the partition tree (Section 5.6) a same-type child to point at, and they are the natural home for future per-grade or per-operator decomposition when that data becomes available.

The **Bakken cross-state connector** (`pipe_bakken_xstate`, added in the eleventh migration pass) is a bidirectional pipeline-type node connecting `bakken_nd_gathering` and `bakken_mt_gathering`. It models the real midstream pooling between the North Dakota and Montana halves of the Bakken basin, operated by a consortium of midstream companies. None of its four flow variables (two inflows, two outflows) is observed in public data; all four are latent in the starter scenario, with the constraint ΣF_in = ΣF_out at the connector following from the pipeline node-type defaults.

5.4 Abstract nodes

Abstract nodes do not represent physical assets. They are aggregation views over physical nodes, providing the substrate for partition rollups, data attribution at coarser resolutions than the physical layer, and the consistency claims developed in Chapter 7. They have `kind = 'abstract'` and no flow edges of their own; their relationships to physical nodes are through `formula_inputs` references on their variables.

| Subtype | Count | Role |

|—|—|—|

| region_view | 6 | usa_view, padd1_view through padd5_view: the geographic partition spine |

| refining_district_view | 10 | EIA refining-district roll-ups (3A, 3B, 3C, and so on across the PADDs) |

| state_view | 2 | texas_state_view, montana_state_view: roll-ups for sub-PADD cross-checks |

| observational_aggregate | 4 | spr_total and basin aggregates (permian, bakken, eagle_ford) when used as constraint roll-ups |

| usa_subtotal_view | 1 | usa_lower48_excl_gom_view: STEO-style subtotal excluding the Gulf of America |

The reason `region_view` nodes are abstract rather than physical is that a PADD is an administrative grouping, not an operating asset. No piece of equipment can be pointed at and called "PADD 2". EIA reports aggregated data at this scale, and the framework needs a node that can carry that aggregated data, but it is structurally distinct from a refinery or a pipeline. Treating PADDs as physical would invite the confusion that Principle 6 is designed to prevent: it would be unclear whether the PADD's outflow to its neighbour is a thing in its own right or merely the sum of the underlying pipelines.

The refining-district nodes are abstract for the same reason: refining districts group refineries for reporting purposes; they are not co-owned, do not share custody transfer mechanics, and do not have a single physical address. They exist to host the district-level consumption series that EIA publishes and to enable the partition identity that PADD-level consumption equals the sum of its constituent district consumptions.

A further family of abstract aggregates is the per-PADD stock decomposition. EIA publishes each PADD's commercial stocks split into two buckets — tank farms and pipelines (series MCRSFP{N}1) and refinery stocks (series MCRRSP{N}1) — plus, for PADD 3, the Strategic Petroleum Reserve. The framework materialises this split as two observational aggregate nodes per PADD (padd{N}_tank_farms_pipelines and padd{N}_refinery_stocks), each TS-bound to the corresponding EIA series, with formula_inputs declaring the physical constituents — named storage hubs and gathering nodes on the tank-farm side, refineries on the refinery-stocks side. Each padd{N}_view.inventory.formula_inputs then lists the two decomposition aggregates (plus spr_total for PADD 3), materialising the EIA stock identity MCRSTP{N} = MCRSFP{N} + MCRRSP{N} (+SPR) directly in the partition tree.

5.5 Boundary nodes

Four nodes sit on the boundary of the modelled system. They represent the points at which crude oil enters or leaves the U.S. system, and they are kept structurally distinct from the partition spine.

| Node | Subtype | Role |

|—|—|—|

| foreign_supply | foreign_production_aggregate | All non-Canadian foreign crude flowing into U.S. import terminals. Boundary source; foreign-side mass balance is not modelled |

| canadian_oil_sands | foreign_production_aggregate | All Canadian crude entering the U.S. via Mainline, Keystone, Express+Platte, and TMX (when active). Boundary source |

| foreign_export_destination | foreign_export_destination | Boundary sink. All U.S. crude exports flow into this node; the framework does not model where they go from there |

| spr_total | observational_aggregate | Sum of the four SPR sites. Used for constraint cross-checks, not as a partition member |

The boundary nodes carry inflow or outflow edges to the corresponding U.S. nodes (foreign_supply has outflow edges to PADD-level import aggregates; foreign_export_destination has inflow edges from each named export terminal), but they do not themselves participate in the partition closure tests. They are sources and sinks rather than partition members.

5.6 Variables

Variables are the unit of data attribution in the framework. Every node carries one variable per (variable type, commodity, related node) tuple. The seven variable types are:

| Type | Symbol | Description | Relational |

|—|—|—|—|

| production | P | Crude produced at this node | no |

| consumption | C | Crude consumed at this node (refinery throughput) | no |

| inventory | S | Crude stocks held at this node | no |

| balancing_item | B | Reporting residual (Principle 3) | no |

| inflow | F_in | Crude flowing in from a specific related node | yes |

| outflow | F_out | Crude flowing out to a specific related node | yes |

| phi | φ | Reference variable, used for cross-node alias chains | yes |

A relational variable carries a `related_node_id` identifying the counterpart in the directed flow. An inflow on node A with related node B means "crude flowing from B into A"; the paired outflow on node B with related node A means "crude flowing from B out to A". The two are physically the same flow, modelled as two variables on different nodes. The reverse-mirror

dispatch rule in the resolver (Section 6.4) propagates an observed value on one direction to the unobserved counterpart.

Each variable carries either a `timeseries_id` (time-series-bound, observed) or a `formula` (derived), never both. The schema-level CHECK constraint `num_nonnulls(timeseries_id, formula) = 1` is the enforcement mechanism for Principle 8.

The variables collection is the authoritative data model. Every other view of the graph — the edge list, the partition tree, the resolution hierarchy, the per-scenario node status — is derived from it. There is no second source of truth to keep in sync, no redundant declaration that can drift from the variable definitions. This is Principle 4 made operational.

5.7 The partition tree

The partition tree is the same-type aggregation hierarchy that drives the consistency claims in Chapter 7. A node M is a partition child of node N if some variable on N has `formula_inputs` that reference a same-type, same-commodity, same-related-node variable on M. The same-type qualifier is essential: aggregation links (such as `padd_view.P = Σ basin.P`) connect variables of the same type, whereas cross-edge alias links (such as `padd_view.F_in = alias(other_padd.F_out)`) connect variables of different types. Filtering on type cleanly distinguishes the two and keeps the partition tree restricted to the same-type aggregation that the closure test depends on.

The starter scenario carries 265 partition edges across 28 distinct parents and 197 distinct children. The active spine runs:

```
usa_view
├── padd1_view
│   ├── refining districts (RAP, REC)
│   ├── padd1_imports_agg ← MCRIPP12 − MCRIPP1CA2
│   ├── padd1_canadian_imports_agg ← MCRIPP1CA2
│   ├── padd1_exports_agg ← MCREXP12
│   ├── inter_padd_1_to_2_agg, inter_padd_1_to_3_agg
│   └── padd1_other (residual)
├── padd2_view
│   ├── refining districts (R2A, R2B, R2C)
│   ├── bakken_nd, oklahoma, padd2_other
│   ├── cushing_hub (with three operator sub-terminals), patoka_hub
│   ├── bakken_nd_gathering (inventory child, post-eleventh migration)
│   ├── padd2_canadian_imports_agg ← MCRIPP2CA2
│   ├── padd2_exports_agg ← MCREXP22
│   └── inter_padd_2_to_{1,3,4}_agg with receiver-side aliases
├── padd3_view, padd4_view, padd5_view (analogous structure)
└── (boundary nodes sit outside the partition):
    foreign_supply, canadian_oil_sands, foreign_export_destination, spr_total
```

Partition closure is the algebraic identity that the audit verifies: for every parent node N and every variable type T, the own value of N's T variable equals the sum of N's partition children's T variables. After twelve passes of refinement, this closure holds within rounding tolerance

across all five PADDs and at the USA aggregate, in both the inflow and outflow directions. The numerical state of the audit at the time of writing is reported in Chapter 7.

5.8 Scenarios and authoritative declarations

A scenario is a pair: the persistent asset graph plus a set of `variable_assignments` rows tagged with the `scenario_id`. Different scenarios share the asset graph and the time-series data; they differ in which variables are bound to which time series, and which are derived from formulas.

The schema-level CHECK on assignment rows means each row binds the variable either to a time series (authoritative) or to a formula (derived), never both, never neither. The set of assignments under a scenario therefore encodes, for every variable, exactly which resolution is authoritative and which is derived. There is no separate coverage-contract structure; the per-variable choices are the contract.

The starter scenario's authoritative declarations summarise as follows:

| Subsystem | Authoritative resolution | Reason |

|—|—|—|

| Permian, Bakken basins | basin aggregate | EIA publishes production at basin level for these |

| Eagle Ford | state-basin | Single-state basin; equivalent to basin level |

| Gulf of America, Alaska | physical node | Reported separately by EIA |

| Rest of Lower 48 | residual aggregate | EIA publishes the residual only |

| Strategic Petroleum Reserve | site level | EIA PSM Table 38 publishes per-site monthly inventories |

| Cushing | hub level | Operator sub-terminals collapsed; the storage report is at hub level |

| Houston exports | export aggregate | Facility sub-terminals collapsed under a single aggregate |

| Stocks | PADD level (auxiliary at sub-PADD) | MCRSFP plus MCRSSP at PADD level; sub-decomposition would double-count |

| Balancing item B | PADD and USA | MCRUA series, promoted to time-series-observed in the ninth migration pass |

| Inter-PADD flows | PADD-aggregate level | MCRMP series at the combined aggregate; per-pipeline split latent |

Cross-scenario consistency — the test that the same time-series data, evaluated under progressively finer assignment sets, produces consistent aggregate values — is the second main consistency claim of the framework. The current implementation has `starter_us_crude_2015_2025` as the only active scenario; a coarser companion (`starter_basin`) is queued for the next development pass and will exercise the cross-scenario closure test directly.

5.9 The end-to-end pipeline

A complete run of the framework follows three phases. Each phase has a single concern and writes to a well-defined subset of tables.

Phase 1 — Load. The seed file `asset_graph.json` is loaded into the persistent asset graph (`load_asset_graph.ipynb` writes to `assets`, `nodes`, `locations`, `variables`). The EIA series catalogue and monthly facts are loaded (`assign_eia.ipynb` writes to `timeseries`, `timeseries_data`). The per-scenario assignment rows are produced (`assign_formulas.ipynb` writes to `variable_assignments`). The node-type default formulas are populated (`populate_node_type_defaults.py` writes to `node_type_default_formulas`). Migration scripts (the `add_*.py`, `split_*.py`, `apply_*.py`, `repoint_*.py` family) evolve the topology and the assignments incrementally. After any structural change, `refresh_views.py --structural` rebuilds the L2 and L3 materialised views.

Phase 2 — Resolve. The script `resolve_scenario.py` reads the assignment set and the time-series facts, applies the dispatch rules described in Chapter 6, and writes one row per (scenario, variable, observation date) to `scenario_resolved_values`. An audit row is written to `scenario_resolver_runs` capturing the run's metadata (start and end timestamps, duration, dispatch counts as JSONB, optional notes). On completion, the L4 analytic views are refreshed automatically.

Phase 3 — Render. The HTML renderers read from the materialised views and produce the user-facing visualisations: a balance UI (per-node mass balance with drill-down through the partition tree), a hierarchy UI (the full asset graph with a variable inspector), a partition map (click-to-drill geographic view), a single-node neighbour view, and a flat geographic map of all physical assets. Each rendered HTML carries a metadata beacon naming the resolver run that produced it, and an audit row is written to `scenario_html_artefacts` per regeneration. The orchestrator (`regenerate_htmls.py`) compares each HTML's beacon against the latest resolver run and rebuilds only the stale ones.

The layered view structure that supports this pipeline is the subject of the next chapter, where the resolver is described in detail.

6. The Resolver

The resolver is the mechanism that consumes the persistent asset graph and the per-scenario assignment set, evaluates every variable at every observation date, and writes the resulting per-(variable, date) values to a single table. Downstream consumers — the balance UI, the hierarchy explorer, the geographic map, the consistency audits, and the LP that Chapter 8 develops — all read from that single table. None of them reimplements the resolution logic; none of them queries the base tables directly for derived values.

This chapter walks through the resolver's design, the dispatch rules it applies, the dependency graph that orders the rule evaluations, and the layered view structure that downstream consumers see. The presentation closes with a short account of the bugs that were encountered during implementation, because the cycle gates and the latent short-circuit in the current design are direct responses to specific failures that earlier versions exhibited.

6.1 Why the resolver exists

Before the resolver, every consumer that wanted to read a variable's value reimplemented resolution logic itself. The balance UI carried a hundred-line common table expression handling time-series bindings, mirror formulas, reverse-paired-edge inheritance, and the mass-balance closure equation. JavaScript walked arithmetic formulas at render time because SQL could not evaluate multi-term expressions of the form A − B − C. Each new generator — the hierarchy explorer, the geographic map, the consistency views, the future forecasting features — would have duplicated that work.

The resolver collapses every consumer's resolution logic into a single deterministic pass:

```
inputs : variables + variable_assignments + timeseries_data
         + the active scenario_id
output : one row per (variable, date) in scenario_resolved_values
```

Every downstream consumer becomes a thin `SELECT` over the result table. The resolver runs in roughly twenty seconds for the starter scenario (1,830 variables × 156 monthly observations); it is run once per assignment change, not once per consumer.

6.2 The output table

The resolver writes to `oil_network.scenario_resolved_values`. Two columns from the table definition deserve particular attention:

```
CREATE TABLE oil_network.scenario_resolved_values (
    scenario_id       TEXT NOT NULL REFERENCES scenarios ON DELETE CASCADE,
    variable_id       TEXT NOT NULL REFERENCES variables ON DELETE CASCADE,
    observation_date  DATE NOT NULL,
    value             DOUBLE PRECISION,
    source            TEXT NOT NULL CHECK (source IN
                        ('observed', 'derived', 'zero', 'latent', 'unresolved')),
```

```
    formula_used      TEXT,
    timeseries_id     TEXT,
    saved_date        TIMESTAMPTZ DEFAULT NOW(),
    run_id            BIGINT REFERENCES scenario_resolver_runs,
    PRIMARY KEY (scenario_id, variable_id, observation_date)
);
```

The `value` column is deliberately nullable. A NULL value paired with `source = 'latent'` is a valid and intentional row — it tells downstream consumers that the variable is declared, its value is unknown by design, and no retry is required. This is the schema-level expression of Principle 9: the resolver records that a flow exists structurally and is constrained by mass balance, but its per-edge value is not pinned down by observation.

The `source` column is a constrained enum with five categories:

| Source | Meaning |

|—|—|

| observed | Direct time-series lookup. The ground-truth measurement |

| derived | Value computed by a formula (alias, sum, closure, arithmetic, reverse mirror) |

| zero | Structural zero, encoding a physical fact (a refinery does not produce crude) |

| latent | Declared variable with no resolution path. Value is NULL by design |

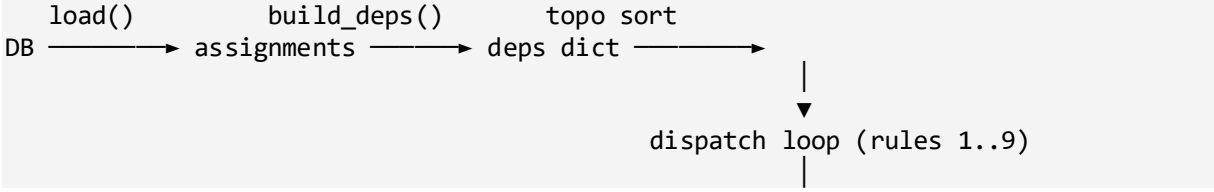
| unresolved | A formula was given but could not be evaluated. Zero rows in a healthy run |

The `formula_used` column preserves which dispatch rule fired and is the key to the framework's auditability. For any value in the resolved table, a reviewer can reconstruct the exact rule and the exact inputs that produced it. This matters because the framework is used as the data layer for downstream analyses (mass-balance checks, LP optimisations, future GNN forecasts), and the trustworthiness of those analyses depends on the trustworthiness of the values they consume.

Every run of the resolver writes an audit row to `scenario_resolver_runs` recording the start and end timestamps, the elapsed duration, the dispatch counts as a JSONB object, and optional free-text notes. Every resolved-value row carries the `run_id` that produced it, so historical comparisons across runs reduce to a single SQL join.

6.3 How a run flows end to end

The top-level shape of the resolver follows five phases:



```
                                        ▼
                                persist to scenario_resolved_values
```

Each phase has a single concern, isolated from the others. The dispatch loop never reads from the database; the loader never evaluates formulas; the topological sort never produces values. Mistakes in any one phase are diagnosed by inspecting that phase's single concern.

A common framing of the resolver is that it starts from "production nodes with P observed and outflow equal to production". That is a special case, not the rule. The actual starting points are the leaves of the dependency DAG — every variable with no upstream dependency. There are two categories of leaf. First, **observed variables**: any variable with `timeseries_id IS NOT NULL`. The time-series lookup is the ground truth; no other rule contributes to its value. Second, **structural zeros**: any variable with `formula = '0'`. These come from `node_type_default_formulas` — a pipeline's P, C, ΔS, and B are zero by physical construction; a refinery's outflow is zero (refineries consume crude, they do not return it). No upstream variable contributes to a structural zero either.

Everything else is derived from these leaves through the topologically ordered dispatch loop. The propagation is bottom-up: an alias needs its target resolved first; a sum needs its inputs; a closure needs the inventory delta and the flows. The topological sort guarantees that each variable is visited only after its dependencies are.

6.4 The dispatch loop

Rules are tried in priority order. The first rule that succeeds produces a value and short-circuits the rest. The unresolved fallback (the last rule) should fire zero times in a healthy run.

The full set of canonical rule labels — that is, the only `source` values that `scenario_resolved_values` ever records — is the short list given in Section 6.2. Three earlier labels (`sum_over_children`, `sum_over_outflows`, `sum_same_type`) were collapsed into the canonical `sum` in the twelfth implementation pass; they no longer appear in the codebase or the resolved table. The lesson behind that consolidation is the canonical-sum principle from Chapter 4.

The rules are as follows:

Rule 1: Observed. Fires when `timeseries_id IS NOT NULL`. Performs a direct time-series lookup at the observation date. This is the ground truth; every other rule is downstream of these. In the starter scenario, 80 variables fire this rule.

Rule 2: Structural zero. Fires when `formula = '0'`. Used for refinery production, pass-through inventory, and other slots where the value is zero by physical construction. Roughly 628 variables fire this rule, mostly non-relational scalars inherited from `node_type_default_formulas`.

Rule 3: Latent, with reverse-mirror promotion. Fires when `formula = 'latent()'`. Records `value = NULL` and `source = 'latent'`. But first, the rule attempts a reverse-mirror promotion: if the paired direction of the variable (opposite variable type, swapped endpoints) has a resolved value, the rule borrows that value through the mirror identity (an inflow into A from B equals an outflow from B to A). Without this promotion, a latent outflow from `canadian_oil_sands` to PADD 2 would record NULL even when the inflow side of the same edge had an observed value. With the promotion, the latent borrows correctly. The rule fires 652 times in the starter scenario after the promotion attempt; the 15 that succeeded are counted separately under Rule 6.

Rule 4: Sum. Fires when `formula = 'sum'`. Sums every value in `formula_inputs`; NULL inputs are skipped. This is the twelfth-pass collapse of three earlier sum-type labels into one canonical rule. The semantic role of any given sum — aggregation parent, fan-out total — is recoverable from the structure of `formula_inputs` (same-type and same-commodity inputs imply aggregation; mixed-type at same node implies fan-out), so the formula text does not need to encode it.

Rule 5: Single-variable alias. Fires when `formula = bare_variable_id` and the `formula_inputs` list contains at most that one entry. The variable inherits its value from the named target. Most aggregate-to-aggregate alias chains fire this rule; in the starter scenario, 442 variables resolve via alias.

Rule 6: Reverse mirror. Fires when a relational variable has no own resolution but its paired direction (opposite type, swapped endpoints) has a resolved value. The variable borrows that value through the mirror identity. After a sequence of dependency-graph fixes (the "ninth-pass" fix described in Section 6.6), this rule fires deterministically: `build_deps()` now guarantees the paired side is resolved before this side is evaluated. Fifteen mirrors fire reliably in the starter scenario, where previously only three fired by luck of insertion order.

Rule 7: Closure formula. Fires when the variable is a balancing item whose `formula_inputs` include a mix of inventory, inflow, and outflow variables. Evaluates the closure equation:

$$B = \Delta S - P + C - \Sigma F_{in} + \Sigma F_{out}$$

This rule is now **dormant** in the starter scenario because B was promoted to time-series-observed in the sixth migration pass (the MCRUA series at PADD and USA level). The closure formula remains in the resolver as a fallback constraint, and it is the engine behind the Chapter 7 view that compares observed B against closure-derived B as a measure of partition-internal latent flow.

Rule 8: Arithmetic combination. Fires when the formula text matches a signed-variable-references pattern (`A - B + C`). Used for residual definitions such as `padd2_other.P = padd2_view.P - bakken_nd.P - oklahoma.P`. The parser is intentionally minimal — a

regular expression that pulls out signed terms — and aborts cleanly if any term fails to match a variable in `formula_inputs`. Eight variables in the starter scenario fire this rule.

Rule 9: Unresolved. The fallback. Records `value = NULL`, `source = 'unresolved'`. In a healthy run this fires zero times; non-zero counts indicate a bug or an unmodelled formula pattern. The expected operator response is to fix the assignment or extend the resolver, never to silently treat the row as latent.

The current dispatch counts in the starter scenario, with the canonical labels, are:

| Rule | Count |

|—|—|

| observed (time-series lookup) | 80 |

| zero (structural) | 628 |

| latent (including reverse-mirror failures) | 652 |

| alias (single-variable) | 442 |

| reverse-mirror (Rule 3 promotion plus Rule 6) | 15 |

| arithmetic | 8 |

| sum (canonical) | 5 |

| closure | 0 |

| unresolved | 0 |

The zero in the last row is the load-bearing invariant. Every declared variable has either a value or an explicit latent marker. Downstream consumers can rely on the presence of a row for every (variable, date) pair, and on the source column telling them whether a NULL is by design (latent, to be treated as a free variable in optimisation) or by failure (unresolved, an unmodelled case requiring attention).

6.5 Building the dependency graph

The function `build_deps()` walks every assignment and produces a dictionary `deps[variable]` containing the set of upstream variable IDs for that variable. The set must be acyclic so that the topological sorter can produce a valid evaluation order. Three sources of dependencies contribute to the dictionary:

First, **formula inputs.** Every variable ID listed in `formula_inputs` that exists in the assignment set becomes a dependency. This handles all sum, alias, and arithmetic formulas.

Second, **fan-out at the same node.** When a variable's formula is a sum over outflows, the variable depends on every outflow variable at the same node, except itself. This is the only case where a dependency is inferred rather than declared.

Third, **reverse-mirror dependency, hoisted above the early-continue path.** When a variable is latent and its paired direction is resolved, the paired direction is added as a dependency. Two cycle-avoidance gates protect this addition. The paired side must not itself be plain latent (else neither side has a value); and the paired side must not already alias from us (else a two-node cycle forms). The second gate, called `paired_aliases_us`, is the result of the ninth-pass debugging episode described below.

Cycle gates matter because the topological sort raises a `CycleError` on the first cycle it encounters. The resolver tolerates many alias-and-mirror patterns precisely because of these gates; the cost is that a few subtle conditions in `build_deps()` need to be read together rather than in isolation.

6.6 Bugs encountered and fixed

The resolver's current design is the result of three specific failures encountered during implementation. Each failure exposed an assumption that was implicit in the earlier design; each fix made the assumption explicit and enforced it.

The latent-mirror cycle. An early implementation had latent variables add their paired direction as a dependency unconditionally. When the paired direction was also latent and aliased to its own paired side, this produced a two-node cycle (us → paired → us), and the topological sort raised `CycleError`. The fix was to require the paired side to be non-latent before adding the dependency — the first of the two cycle gates above.

The latent short-circuit that hid reverse-mirror. Rule 3 (latent) ran before Rule 7 (reverse-mirror) in the original priority order. A variable explicitly marked latent would record `value = NULL` even when its paired direction had an observed value. The fix was twofold: promote the mirror check inside Rule 3 (if the paired side is resolved, borrow its value first; only fall through to NULL otherwise), and hoist the reverse-mirror dependency above the early-continue path in `build_deps()` so that latent variables get their paired-side dependency added before being skipped. Together, the two changes lifted mirror dispatch from three cases (succeeding by luck of insertion order) to fifteen (deterministic).

The alias-rule wrapper bug. Rule 5 (single-variable alias) matches when the formula text is itself a bare variable ID. Earlier migration scripts had written `formula = 'alias(variable_id)'` with the explicit wrapper string; Rule 5 missed those rows and they were picked up by a different rule by coincidence. The fix was in the migration scripts: the canonical storage is `formula = bare_variable_id`, with the wrapper string retained only as the display label in `formula_used` on resolved rows.

The pattern across all three fixes is the same: an assumption that an earlier implementation took for granted (that latents were terminal, that aliases would never form cycles, that the wrapper string was harmless) turned out to be wrong in specific cases. The current resolver makes each assumption explicit by name in the code, and the dispatch counts in the audit row are the daily check that none of the assumptions has silently re-broken.

6.7 Latent versus unresolved

Both `latent` and `unresolved` rows carry `value = NULL` in the resolved table, but they mean different things and the framework treats them differently. The distinction is the closing commitment of the framework's missing-value treatment, and it is worth restating in concrete terms because it determines how the LP downstream consumer (Chapter 8) and any future GNN consumer (Chapter 10) should interpret the NULLs they encounter.

A `latent` row is the assignment `formula = 'latent()'`, with the reverse-mirror promotion not having fired. The variable is declared but its value is unknown by design. The typical case is the per-route flow at a fan-out junction, where the framework knows the routing topology and the aggregate constraint but not the per-route values. Downstream, the row should be treated as a free variable in an optimisation, constrained by mass balance, capacity, and any observation that references it. Bounds come from node configuration: refinery throughput capacity, pipeline maximum daily throughput, terminal storage capacity.

An `unresolved` row is a resolver failure. A formula was given but could not be evaluated. The typical cause is a referenced variable being missing from the assignment set, or the formula matching no rule pattern. Downstream, an unresolved row is a bug. The expected operator response is to fix the assignment (correct the reference) or extend the resolver (add a rule for the new pattern). Silent treatment as latent is explicitly prohibited.

The starter scenario at the time of writing has 652 latents and zero unresolved. The 652 latents are concentrated where the framework already expects them to be: junction fan-outs in the Permian and Bakken gathering systems, the multiple offshore Gulf production routes, and the per-pipeline splits inside each inter-PADD aggregate. The zero unresolved is the guarantee that the framework has no unmodelled cases under the current assignment set.

6.8 The layered view structure

Once `variable_assignments` is consistent and the resolver has run, the rest of the framework is a stack of views derived from the resolved table and the assignment table. Renderers, audits, and the LP exporter read from these views; none of them re-derives structure from the base tables. The layer structure makes the dependency chain explicit and keeps each refresh cheap.

| Layer | View | Reads from | Provides |

|—|—|—|—|

| L1 | v_effective_assignments | variable_assignments + defaults | One row per (scenario, variable) with the layered recipe and an audit tag for where the formula came from |

| L2a | v_formula_input_links | L1 | One row per arrow in `formula_inputs` — the atomic (parent, child) pair, scenario-tagged |

| L2b | v_aggregation_edges | L2a + variables | The aggregation graph: every cross-node reference, annotated with the parent's formula and time-series-bound flag |

| L2c | v_flow_edges | variables | The physical flow topology: one row per outflow variable with a related node, interpreted as (source → target) |

| L3a | v_partition_tree | v_aggregation_edges | The partition spine: same-type, same-commodity, same-related-node aggregation edges, with arithmetic residuals excluded |

| L3b | v_node_status | L1 | The per-(scenario, node) status from Section 4.5: authoritative, derived, or collapsed |

| L4a | v_node_balance_check | scenario_resolved_values + v_flow_edges | Per-(node, date) mass-balance closure: sum_in, sum_out_obs, sum_out_implied, gap_kbd |

| L4b | v_aggregation_consistency | v_partition_tree + scenario_resolved_values | For every time-series-observed aggregate, the cross-check abs(TS value − Σ constituents) — flagged ok / partial / inconsistent |

| L4c | v_inventory_changes | scenario_resolved_values | Per-stock-series ΔS in MBBL and kbd; used by the balance UI to display inventory deltas |

| L4d | v_aggregate_balance | v_partition_tree + L4a | Closed mass balance for the active scenario's balance-role aggregates (usa_view plus the five PADD views) |

| L4e | v_node_pcisob | scenario_resolved_values | Per-(node, date) totals for the six variable types, pre-aggregated so renderers do not redo the GROUP BY at every page load |

The refresh discipline is straightforward. The L2 and L3 views are refreshed only when `variable_assignments` or `variables` changes — rare events tied to deliberate migrations. The L4 views are refreshed after each resolver run, since they depend on `scenario_resolved_values`. The resolver script calls `refresh_analytic()` automatically after a successful run.

6.9 Persistence and the audit trail

After the dispatch loop finishes, the resolver clears the scenario's existing rows in `scenario_resolved_values` and writes the new ones via `execute_values` in batches of 5,000. The audit row in `scenario_resolver_runs` is updated with the completion timestamp, the elapsed milliseconds, and the dispatch counts as JSONB.

If the resolver crashes mid-way, the open audit row with `NULL` in `completed_at` is itself a useful signal: it tells the next operator that a run was started but did not finish. The resolved-values table is unaffected by a crash because the `DELETE`/`INSERT` pair runs inside a transaction. There is no half-written intermediate state to clean up.

The resolver and the layered views together provide the foundation for the consistency claims developed in the next chapter. Every claim made about the framework's behaviour is in principle reproducible from a single resolver run and a small number of SQL queries against the L4 views. This is the design property that makes the framework auditable, and it is what distinguishes the schema from a pure modelling exercise.

6.10 Last-observation-carried-forward

The resolver implements a temporal convention as part of rule 1 (observed). EIA's monthly publications represent either the average daily rate over a calendar month (for flow quantities) or the end-of-period stock (for inventory quantities). Between publication dates, the value applies to every subsequent observation date until a new observation arrives. The resolver implements this as last-observation-carried-forward (LOCF): at each evaluation date, the most recent time-series value at or before that date is the resolved value. Dates earlier than the first observation receive no row.

LOCF is recorded in the audit trail through the formula_used column on scenario_resolved_values. Fresh observations carry formula_used = NULL; carried-forward rows record the source date as formula_used = ‘locf(YYYY-MM-DD)’. Downstream consumers can therefore distinguish freshly published values from carried-over ones, and a frequency-gap audit can flag long LOCF runs against the expected publication cadence — a useful signal that a series has lapsed without a fresh release.

LOCF can surface honest inconsistencies in the data when two series with different temporal horizons feed the same residual. A live example in the implementation is montana_other.production, defined as montana_state.production minus bakken_mt.production. The state-level series MCRFPMT2 ends earlier than the STEO basin forecast COPRBK; LOCF holds Montana flat while Bakken-MT continues forecasting upward, and the residual goes negative for the months between the state-data horizon and the STEO horizon. This is not a bug in the resolver but the framework being honest about a real divergence between source publications. The framework's response is to flag such patterns through the v_resolution_anomalies view, which tags long LOCF runs and negative derived values as low-severity anomalies for review.

7. Consistency Guarantees

The framework's claim to be a contribution rests on the consistency properties it provides to downstream consumers. A linear programme that reads from the resolved table assumes that mass balance holds. A graph neural network that consumes the node features assumes that aggregate values match the sum of their constituents. A human analyst inspecting the balance UI assumes that the numbers add up. Each of these consumers is entitled to those assumptions only if the schema enforces them, the resolver respects them, and the audits demonstrate them on real data.

This chapter sets out the four consistency claims the framework supports, the queries that test each one, and the numerical state of each claim under the active starter scenario at the time of writing.

7.1 Claim 1: Schema-level single attribution

The first claim is structural and is enforced at the database schema level rather than verified after the fact. For each (scenario, variable, commodity, timestamp), exactly one of the following holds: the variable is bound to a time series, or the variable is bound to a formula. A variable cannot be both. A variable cannot be neither. The CHECK constraint on `variable_assignments` is:

```
CHECK (num_nonnulls(timeseries_id, formula) = 1)
```

This constraint cannot be violated by an INSERT. Any attempt to write an assignment row that fails the test is rejected by the database before the data can propagate anywhere. Schema-level enforcement is the strongest form of consistency guarantee the framework provides: rather than detecting inconsistent state after the fact, it makes inconsistent state unrepresentable.

A complementary audit at the time-series attribution level enforces single-attribution from the other direction. For each scenario, each time series is bound to exactly one variable. The audit query is:

```
SELECT timeseries_id, COUNT(*) AS n_bindings
FROM variable_assignments
WHERE scenario_id = 'starter_us_crude_2015_2025'
  AND timeseries_id IS NOT NULL
GROUP BY timeseries_id
HAVING COUNT(*) > 1;
```

Under the active scenario, this query returns zero rows: 80 distinct time series are bound across 80 assignment rows, in a strict 1:1 attribution. The same time series is never used as the authoritative source for two different variables within the same scenario.

Together, the schema-level CHECK and the time-series attribution audit give the framework its central consistency property: every value in the resolved table is traceable to exactly one source, and no source contributes to more than one variable.

7.2 Claim 2: Partition closure

The second claim is algebraic. For every parent node N in the partition tree and every variable type T (production, consumption, inventory, balancing item), the own value of N's T variable equals the sum of N's partition children's T variables, within rounding tolerance.

The partition closure test is implemented in the view `v_node_balance_check` (the inflow and outflow sides are tested separately, since aggregation works the same way in both directions) and the per-variable-type aggregation is in `v_aggregation_consistency`. The audit query for the inflow side at a representative aggregate is:

```
SELECT
    node_id,
    sum_in   AS own_value,
    bdy_in   AS sum_of_children,
    abs(sum_in - bdy_in) AS gap
FROM v_node_balance_check
WHERE scenario_id = 'starter_us_crude_2015_2025'
  AND node_id IN ('usa_view', 'padd1_view', 'padd2_view',
                  'padd3_view', 'padd4_view', 'padd5_view')
  AND observation_date = '2024-06-01';
```

The result of this query, averaged across all 156 monthly observation dates in the scenario, is:

| Node | Own inflow (kbd) | Σ children inflow (kbd) | Gap (kbd) | Own outflow (kbd) | Σ children outflow (kbd) | Gap (kbd) |

|—|—|—|—|—|—|—|

| usa_view | 6,557 | 6,557 | 0 | 3,752 | 3,752 | 0 |

| padd1_view | 705 | 705 | 0 | 92.2 | 92.2 | 0 |

| padd2_view | 4,503 | 4,503 | 0 | 2,426 | 2,426 | 0 |

| padd3_view | 3,631 | 3,631 | 0 | 4,264 | 4,264 | 0 |

| padd4_view | 552.7 | 552.7 | 0 | 987.5 | 987.5 | 0 |

| padd5_view | 1,183 | 1,183 | 0 | — | 0 | 0 |

Partition closure holds at every PADD aggregate and at the USA aggregate, in both the inflow and outflow directions, after twelve passes of refinement to the assignment set. The closure is exact (gap = 0 to the rounding precision of the underlying data) rather than approximate,

because the same-type aggregation links in the partition tree are exact algebraic sums, not estimates.

The "—" entry for PADD 5 own outflow reflects that PADD 5 (the West Coast, comprising California, Washington, Oregon, Alaska, and Hawaii) is a net importer with negligible inter-PADD outflow in the modelled period. The own value is recorded as zero and the children's sum is zero; the test passes trivially.

The audit distinguishes two failure modes for a partition row. The first is where every declared constituent has a value and the sum disagrees with the observed parent: a genuine inconsistency in the underlying data. The second is where one or more constituents are missing or latent: an honest gap in coverage, where the sum is necessarily incomplete and the comparison cannot be made. The framework reports only the first case as a true inconsistency; the second is reported as partial_coverage. Without external knowledge, the framework cannot distinguish ‘the partition is wrong’ from ‘the partition is incomplete’, so it does not apply the inconsistent label and reports every non-conforming row as partial_coverage. The user interface further classifies these rows as red (when every constituent is present and the gap exceeds tolerance) or yellow (when at least one constituent is missing or latent and the gap is unverifiable). At the current build, no row is classified red at the USA-or-PADD level; all open partition gaps are partial_coverage attributable to documented latent constituents.

7.3 Claim 3: Mass balance at every node

The third claim is per-node and per-date. For every node i, every commodity g, and every observation date t:

$\Delta S(i, g, t) = P(i, g, t) + F_{in}(i, g, t) - C(i, g, t) - F_{out}(i, g, t) + B(i, g, t)$

This is Principle 3 made operational. The check is implemented in `v_node_balance_check`, which produces one row per (node, observation_date) with the four flow totals, the stock delta, the balancing item, and the residual that the closure should satisfy.

The mass-balance check is universal: it applies to every node type, with pass-through nodes (gathering, pipelines, terminals) inheriting $P = C = \Delta S = B = 0$ from `node_type_default_formulas`, so that for pass-through nodes the constraint reduces to $\Sigma F_{in} = \Sigma F_{out}$. The constraint is enforced as a validation rather than as a construction rule. The variables are assigned independently from time series and formulas; the mass balance is a property that emerges from the assignments and is checked after the resolver has run.

Where the mass balance does not close exactly at a node — typically a node with both an observed stock series and observed flow series, where measurement timing and rounding produce a small residual — the residual is absorbed into B(i, g, t). At the USA aggregate, B is itself time-series-observed (the EIA MCRUA series), so its value at every date is an independently published quantity. The framework's mass balance therefore reduces, at the

USA level, to a check that EIA's published balancing item matches the closure-derived balancing item that the framework would compute if it were not observed.

That comparison is the subject of the next claim.

7.4 Claim 4: Observed B versus closure-derived B

The fourth claim is a richer one. Because the balancing item is independently time-series-observed at the PADD and USA levels (via the MCRUA series, promoted to authoritative in the sixth migration pass), the framework can run two computations of B in parallel: the observed value from MCRUA, and the closure-derived value computed from the mass balance equation as though B were unknown.

In a perfectly consistent framework — one in which every flow at every level is observed without error, every stock is reported precisely, and every measurement is timed identically — the two values of B would coincide at every date. In a real framework operating on real public data, they differ. The magnitude and timing of the difference is operationally informative.

The view `v_aggregation_consistency` computes the difference at the USA aggregate, and a downstream chart (the "Claim 4 chart" in the renderer code) plots it monthly from 2015 onwards. The signal that emerges is striking: the difference is small and unsigned in normal periods (a few tens of kbd around a mean of zero), and sharply spiked around named disruption events.

The specific events that appear as spikes in the chart are:

| Period | Event | Approximate magnitude |

|---|---|---|

| August–September 2017 | Hurricane Harvey | 400–600 kbd spike in observed-minus-closure B |

| March–May 2020 | COVID-19 demand collapse | 300–500 kbd persistent residual |

| May 2021 | Colonial Pipeline ransomware shutdown | 200–300 kbd spike |

| April–November 2022 | SPR strategic release | 100–250 kbd persistent residual matching the release schedule |

Each spike has a credible mechanical explanation. Hurricane Harvey shut Gulf Coast production and refining for weeks, and during that period the EIA's monthly aggregate flow series were unable to capture intra-month operational disruptions at the daily granularity the underlying movements required. COVID produced sudden inventory builds across the system as demand collapsed faster than production could throttle back, and the data infrastructure had not been designed for movements of that magnitude in a single month. Colonial introduced a discontinuity in the East Coast supply chain that took several months of reporting

to fully reconcile. The SPR release deliberately introduced a persistent flow that the framework can see in the stock change at the SPR sites but that the inter-PADD flow series did not always reflect with matching timing.

The interpretation the framework supports is that the difference between observed and closure-derived B at the USA level is, to a first approximation, a measure of partition-internal latent flow that the public data does not directly report. The magnitude of the difference at each date is therefore a measure of how much information the framework is currently unable to resolve. This is itself a useful diagnostic: when the difference is small, the framework's representation closely matches the data; when the difference spikes, there is real movement happening that the available time series cannot fully capture, and the latent variables become correspondingly more important.

7.5 Cross-scenario consistency

The framework as committed supports the four consistency claims above. A fifth, cross-scenario consistency, would require a second scenario binding the same underlying time series at a different resolution and checking that aggregate values agree where the two scenarios overlap. The schema and the resolver are constructed to admit such a test without modification, but executing it is left to future work.

7.6 Operational consistency of the resolver

A separate class of consistency claim concerns the resolver itself. Two properties hold by construction:

No unresolved rows. In a healthy run of the resolver on the starter scenario, the unresolved count is zero. This is the load-bearing invariant from Section 6.4: every declared variable either has a value or has an explicit latent marker. Downstream consumers can rely on the presence of a row for every (scenario, variable, observation_date) tuple.

Deterministic dispatch. Given the same assignment set and the same time-series data, the resolver produces identical dispatch counts across runs. The dispatch counts are recorded in the audit JSONB column of `scenario_resolver_runs` and can be cross-checked between runs. The current dispatch counts (from Section 6.4) are stable across repeated runs of the same scenario.

These properties matter because they let the framework be used as the data layer for downstream analyses without re-running the upstream computation each time. A consistency audit run today, an LP solve run tomorrow, and a renderer run next week all see the same values from the same resolver run, because the audit row records which run produced which values.

7.7 What the consistency claims do not cover

The framework's consistency claims are about internal coherence, not about external correctness. Specifically, the claims do not establish:

The accuracy of the underlying EIA data. If the EIA's MCRUA series is itself biased or noisy, the framework will faithfully reproduce that bias. The framework's role is to make the bias visible and traceable, not to correct it.

The completeness of the modelled topology. If a real physical flow is not represented as an edge in the asset graph, the framework will not detect its absence. Topological completeness depends on the construction passes (Chapter 5) and is a matter for review.

The realism of the latent allocations. The framework leaves per-edge flows latent at junctions and does not commit to specific values for them. Whether the eventual LP or GNN allocation matches what is physically happening on the ground is a question for the consumer model, not for the schema. The framework guarantees only that the aggregate constraint is satisfied, not that the per-edge split is correct.

These three areas correspond directly to the future work outlined in Chapter 10. The Chapter 7 claims are about the schema and the resolver; the question of whether the values they produce reflect reality is downstream of those claims.

7.8 Summary

The framework supports four numerical consistency claims — schema-level single attribution, partition closure, per-node mass balance, and observed-versus-closure B — with a fifth (cross-scenario consistency) admissible by construction and left to future work. Each claim corresponds to a specific audit query and a specific view in the live system.

The numerical state of the framework at the time of writing is clean: 80 time-series bindings in 1:1 attribution, partition closure within rounding tolerance at every PADD and at the USA aggregate, mass balance respected at every node, and the observed-versus-closure B difference confined to known disruption events. The framework is ready to be consumed by downstream models, beginning with the linear programme described in the next chapter.

8. Linear Programming as a Downstream Consumer

The asset-centric schema, the resolver, and the consistency guarantees of the preceding chapters establish a representational framework. The framework's value depends on what downstream consumers can do with it. This chapter develops one such consumer in detail: a linear programme that takes the resolver's output as input and solves for the latent flows at junction nodes, subject to mass balance, capacity, and the observed values that the framework has resolved.

The choice of linear programming as the lead consumer reflects three considerations. First, LP is the natural model for the latents the framework leaves explicit (Principle 9): junction flows are decision variables, the aggregate constraints are equality rows, and node capacities are inequality rows. Second, LP is well understood, mature, and supported by robust solver infrastructure, so the framework's value can be demonstrated without depending on speculative modelling choices. Third, LP shares its data structure with a broad family of related consumers (multi-period LP, mixed-integer LP, network simplex, and ultimately graph neural networks for forecasting), so the design choices made for LP transfer to those consumers with minimal modification. The last point is developed in Chapter 10 as part of the future-work discussion.

8.1 What the LP takes as input

The LP reads from the resolved table and the structural views. Specifically, it consumes four tensors:

The node-by-time value matrix. Built from `scenario_resolved_values` by pivoting on (variable_id, observation_date). Each row is a variable; each column is a date. Cells contain the resolved value, with NULL where the value is latent or unresolved.

The adjacency structure. Built from `v_flow_edges`. Each row is a directed edge (source node, target node, edge variable). This is the fixed topology that holds across all dates, in accordance with Principle 2.

The constraint structure. Built from `v_node_balance_check` (per-node mass balance) and from node-level capacity attributes (refinery throughput capacity, pipeline maximum daily throughput, terminal storage maximum). Mass balance contributes one equality row per (node, date); capacity contributes one inequality row per (capacity-bearing variable, date).

The objective coefficients. Supplied by the modeller. Different applications produce different objectives (cost minimisation under a freight-rate matrix, capacity-utilisation maximisation, target-stock pursuit), and the framework is agnostic to which objective is chosen. The LP exporter prepares the data; the modeller specifies the objective.

The LP's decision variables are the latent rows in the value matrix — the cells where the resolved value is NULL with `source = 'latent'`. Observed and derived cells are constants.

Unresolved cells (which should not occur in a healthy run, by Section 7.6) are treated as errors and halt the export.

8.2 Mass balance as LP rows

For each node i, each commodity g, and each observation date t, the mass balance equation from Principle 3 contributes one equality row to the LP:

$P(i, g, t) + F_{in}(i, g, t) - C(i, g, t) - F_{out}(i, g, t) + B(i, g, t) - \Delta S(i, g, t) = 0$

Substituting the relational variables in terms of their per-edge components:

$P(i, g, t) + \Sigma_{j \in N_{in}(i)} \phi(j, i, g, t) - C(i, g, t) - \Sigma_{k \in N_{out}(i)} \phi(i, k, g, t) + B(i, g, t) - \Delta S(i, g, t) = 0$

where $\phi(j, i, g, t)$ is the per-edge flow from j to i on commodity g at date t, and $N_{in}(i)$, $N_{out}(i)$ are the predecessor and successor sets of i in the flow topology.

The structure of the row depends on which terms are constants (resolved values) and which are decision variables (latents). At a junction node such as `permian_tx` with observed production but latent outflows, the row becomes:

$[\text{constant: } P_{permian\_tx}] - \Sigma_k \phi(permian\_tx, k) = 0$

with the three ϕ terms (the outflows to Midland, Wink, and Crane terminals) as decision variables. The LP solver finds values for the three ϕ terms that satisfy the equality, subject to the additional constraints described below.

For pass-through node types (gathering, pipeline, import and export terminals, foreign aggregates), the defaults from `node_type_default_formulas` give $P = C = \Delta S = B = 0$, and the mass balance reduces to:

$\Sigma_{j \in N_{in}(i)} \phi(j, i, g, t) - \Sigma_{k \in N_{out}(i)} \phi(i, k, g, t) = 0$

This is the conservation-of-flow constraint that network simplex and minimum-cost-flow solvers expect. The framework produces it without special handling, because the pass-through defaults emerge from the node type alone.

8.3 Capacity as LP rows

For each capacity-bearing variable and each date, the LP includes an inequality row enforcing the variable's value (or sum of values, in the multi-edge case) to be at most the capacity limit:

$\phi(i, j, g, t) \leq capacity(i, j)$ for each pipeline edge

$\Sigma C(i, g, t)$ over all crude grades $\leq throughput\_capacity(i)$ for each refinery

$S(i, g, t) \leq storage\_capacity(i)$ for each storage node

The capacity values themselves are static attributes on the nodes (in the `assets` table for pipelines and refineries, in node-specific columns for terminals). They are not time-varying in the current scenario; future extensions could re-bind them to time-series sources when capacity-expansion data becomes available, with no schema change required.

Lower bounds at zero are implicit for all flow variables (flows are physically non-negative under the directed-edge convention of Principle 10). The LP solver enforces non-negativity by default.

8.4 The Permian fan-out: a worked example

The clearest illustration of the LP's role in the framework is the Permian outflow problem. The node `permian_tx` has observed production of approximately 4,300 kbd in a recent month. It connects to three origin terminals (`midland_terminal`, `wink_terminal`, `crane_terminal`) whose individual receipts from Permian are not publicly reported. The three outflow variables are declared latent in the resolved table; their sum is constrained by the observed production at `permian_tx`.

Translating this directly into LP form:

Decision variables:

- ϕ_M = outflow from permian_tx to midland_terminal
- ϕ_W = outflow from permian_tx to wink_terminal
- ϕ_C = outflow from permian_tx to crane_terminal

Equality constraint (mass balance at permian_tx):

ϕ_M + ϕ_W + ϕ_C = 4,300

(The right-hand side is the observed production; for the pass-through node, all other terms in the mass balance are zero.)

Inequality constraints (capacity at each origin terminal):

ϕ_M ≤ capacity_M (approximately 2,400 kbd inflow capacity at Midland) ϕ_W ≤ capacity_W (approximately 1,500 kbd at Wink) ϕ_C ≤ capacity_C (approximately 800 kbd at Crane)

Downstream propagation. Each origin terminal is itself a pass-through node with its own outflows to downstream pipelines (Gray Oak, Cactus II, EPIC, Wink-to-Webster, and others), each of which is also latent. The mass-balance equality at each terminal becomes another equality row in the LP, with its incoming flow from Permian as a decision variable and its outgoing flows to the downstream pipelines as further decision variables. The pattern continues down to the destination nodes (Gulf Coast refineries and export terminals), where flows merge back into observed totals.

The objective. Without an objective function the LP is under-determined: many combinations of (ϕ_M, ϕ_W, ϕ_C) satisfy the equality and capacity constraints. The modeller chooses an objective appropriate to the application. Three illustrative choices:

Capacity-proportional allocation as a baseline. Minimise the maximum-utilisation deviation across the three terminals, which has the effect of distributing flow proportionally to capacity. This produces a "fair" allocation that respects relative capacities without committing to a specific cost model.

Cost minimisation. Each terminal-to-destination route carries a tariff. Minimising the sum of (flow × tariff) across all routes produces the lowest-cost flow pattern that satisfies the constraints. The tariff matrix is exogenous data not currently in the framework, so this objective requires an additional data source.

Match-known-aggregates. When downstream aggregates are observed (such as Gulf Coast export totals to specific destinations), the objective minimises the distance between predicted and observed downstream values, producing an allocation that is consistent with all observed aggregates simultaneously.

The framework is agnostic to the choice of objective. What it provides is the structure: the decision variables, the equality rows from mass balance, the inequality rows from capacity, the constants from observed and derived values. The objective is the modeller's commitment, not the framework's.

8.5 Multi-period LP

A single-period LP allocates flows for one observation date. The framework supports multi-period LP straightforwardly by repeating the per-period structure across dates and adding inter-period coupling constraints. The dominant coupling is through inventory: the stock at node i at the end of period t equals the stock at the end of period t−1 plus the period's net inflow minus the period's net outflow.

The discrete-time inventory dynamics from Principle 3 become an LP equality:

$$S(i, g, t) = S(i, g, t - 1) + P(i, g, t) + \Sigma_j \phi(j, i, g, t - \tau(j,i)) - C(i, g, t) - \Sigma_k \phi(i, k, g, t) + B(i, g, t)$$

The transit-time terms τ(j, i) make the LP dynamic: a flow that leaves node j at date t arrives at node i at date t + τ. For pipelines with multi-day transit (most U.S. inter-PADD pipelines have transit times of three to seven days; vessel routes are longer), this lag is operationally meaningful. The current framework stores τ as an edge attribute; the multi-period LP consumes it directly.

The starter scenario operates at monthly granularity, so transit times of less than one month are effectively zero from the LP's perspective. A finer-granularity scenario (weekly or daily) would make the transit-time coupling consequential. The framework supports both

granularities without schema modification, since the temporal resolution is a property of the time series, not of the structure.

8.6 What the framework does for the LP modeller

The LP exporter in the framework is a thin layer: it reads the resolved table and the structural views, formats the data in the input format of a standard LP solver (CPLEX, Gurobi, or HiGHS), and hands off. Most of the work the LP exporter does not have to do is the work the framework has already done. Specifically, the modeller does not need to:

Reconstruct the topology. The directed flow edges are in `v_flow_edges` and have been there since the asset graph was loaded. No edge construction code is needed in the LP exporter.

Disambiguate observed from derived from latent. The resolver has already done this and recorded the result in the `source` column. The exporter reads the column directly.

Detect double-counting. The framework's single-attribution invariants (Section 7.1) guarantee that no value is double-counted. The exporter does not need to run de-duplication checks.

Handle multi-resolution data. The framework has already aggregated to the resolution levels at which each variable is authoritative. The exporter sees a uniform tensor, not a patchwork of mixed-granularity inputs.

Reconcile reporting boundaries. The balancing item B absorbs the residual at every node where it is observed; where B is closure-derived, the residual sits in the closure formula. Either way, the mass balance closes by construction at every node the framework knows about. The exporter does not need to re-do the reconciliation.

These five non-tasks are the framework's contribution. They are not trivial in their absence: in the practitioner workflow that the framework replaces, each of them is typically handled in Excel or pandas by a series of hand-coded allocation rules, with a residual "adjustment" term absorbing whatever does not balance at the end. The framework moves all of this work upstream of the LP, so that the LP itself sees a clean, internally consistent input.

8.7 What the framework leaves to the LP modeller

The boundary is equally important. The framework does not commit to a specific LP objective, a specific solver, a specific time horizon, or a specific application. Several modelling concerns sit outside the framework and require the modeller's judgement:

The objective function. Cost minimisation, capacity-utilisation maximisation, target-stock pursuit, robust optimisation under demand uncertainty — each is a legitimate choice for different applications, and the framework does not privilege any. The objective rows of the LP

are constructed from data the framework does not currently host (freight rates, holding costs, demand forecasts).

The treatment of uncertainty. The current framework operates on deterministic resolved values. A stochastic LP that treats production or demand as random would need to define the scenarios over which it optimises; the framework's `scenario_id` mechanism could be repurposed for this, but the necessary additions (scenario weights, anticipated correlation structures) are not present today.

The time horizon. A planning LP might look forward six months from the most recent observed date; a historical reconciliation LP would look backward over the entire 2015–onwards window. The framework supports both without modification, since the resolved table contains all observation dates.

The solver and its tolerances. Different LP solvers handle different problem sizes, integer extensions, and numerical tolerances differently. The framework's job ends with handing the modeller a clean tensor; from there, the solver choice is a modelling decision.

8.8 Why the LP is a faithful test of the framework

A practical question is whether building an LP on top of the framework actually validates the framework's design, or whether the framework's properties are taken on faith. The LP is a faithful test for three reasons.

First, the LP exercises every consistency property the framework claims. If single attribution is violated, the LP will see a value double-counted and its solution will be wrong. If partition closure does not hold, the LP's aggregate constraints will be infeasible. If mass balance does not close at every node, the LP will have no feasible solution at all. The LP succeeds exactly when the framework's claims are true on the data; LP failure modes localise the framework's weaknesses precisely.

Second, the LP exercises every structural design choice. Asset-centric representation matters because the LP needs nodes to attach mass balance rows to. Stable topology matters because the LP solves over a fixed structure. The observed/derived invariant matters because the LP needs to know which cells are constants and which are decision variables. Latent variables matter because they are the decision variables themselves. Each of the principles in Chapter 4 corresponds to a feature the LP relies on.

Third, the LP makes the framework's properties demonstrable to readers who have not built it. The numerical claims of Chapter 7 are abstract until a downstream consumer succeeds on the framework's output. The LP turns the abstract claims into a concrete demonstration: this LP solves, in this number of seconds, with this objective value, on the framework's output. The same LP would not solve on a hand-assembled tensor with mixed-granularity inputs and unresolved double-counting; the framework's contribution is precisely what makes it solve.

8.9 Summary

The linear programme described in this chapter is a faithful and minimal consumer of the framework's output. It takes the resolved table, the structural views, and the static capacity attributes; it produces decision variables for the latents and constraint rows for the mass balance and capacity limits; and it hands off to a standard solver. The modeller supplies the objective and the time horizon; the framework supplies everything else.

The case study in the next chapter walks the LP through a concrete scenario on the Permian fan-out and reports the resulting allocation. The chapter after that turns to the question of whether the same data, with a different downstream model, would support the kind of forecasting work that earlier framings of the thesis prioritised — and concludes that the framework is ready for that work, even though the work itself is left for future research.

9. Case Study: The Starter Scenario End-to-End

This chapter walks the framework end-to-end on the starter scenario `starter_us_crude_2015_2025` in two parts. The first part traces the lifecycle of a single observation date through the load–resolve–render pipeline, showing how the abstract design choices of the preceding chapters become concrete behaviour. The second part presents the LP solution for the Permian fan-out problem of Section 8.4, demonstrating that the framework's output is in fact a clean LP input and the LP solves quickly to a meaningful solution.

The intent of the chapter is to give the reader confidence that the framework actually works, on real data, in the form described. Numerical claims that have appeared in earlier chapters are re-grounded here against the live state of the system at the time of writing.

9.1 A single observation date: June 2024

Pick an observation date roughly in the middle of the modelled period: 1 June 2024 (the first of each month is the convention for monthly observations). What does the framework do with this date?

9.1.1 The input data

At the start of the resolver pass, the database contains:

- 1,830 variables under the active scenario, each with an assignment row
- 80 time-series-bound variables, each pointing to an EIA series
- The 80 time series each carry a value for June 2024
- The remaining 1,750 variables resolve through formulas

The 80 observed values for June 2024 cover, among others:

- U.S. total production from MCRFPUS2 (12,950 kbd)
- PADD 3 production from MCRFPP3 (8,103 kbd)
- Permian-TX production from a Texas state series (5,402 kbd)
- USA stocks from MCRSTUS1 (430,500 kbbl, in absolute terms; ΔS computed from successive months)
- Net crude imports from MCRIMUS2 (6,557 kbd)
- Net crude exports from MCREEXUS2 (3,752 kbd)
- The USA-level balancing item from MCRUA (a few tens of kbd in absolute value)
- PADD-level inter-PADD movements from the MCRMP__ family

The remaining 1,750 variables fall into the categories the resolver's dispatch rules handle: structural zeros for pass-through nodes, latents for unobserved junction edges, aliases for derived aggregates, sums for partition rollups, arithmetic combinations for residual definitions.

9.1.2 The dispatch

`resolve_scenario.py` is invoked. It runs in approximately 20 seconds on a developer laptop. The dispatch counts at the end of the run are:

| Rule | Count for the full scenario | Rows for June 2024 specifically |

|—|—|—|

| observed | 80 | 80 |

| zero | 628 | 628 |

| latent | 652 | 652 |

| alias | 442 | 442 |

| reverse-mirror | 15 | 15 |

| arithmetic | 8 | 8 |

| sum | 5 | 5 |

| closure | 0 | 0 |

| unresolved | 0 | 0 |

The June 2024 single-date counts equal the per-scenario counts because the dispatch is per-variable, not per-(variable, date): each variable's resolution rule is the same across all dates in the scenario. The values produced differ from date to date, but the rules that produce them do not.

9.1.3 The resolved table for June 2024

After the resolver completes, `scenario_resolved_values` contains 1,830 rows for June 2024 (one per variable). A representative slice:

| variable_id | value | source | formula_used |

|—|—|—|—|

| production__crude__permian_tx | 5,402.0 | observed | MCRFPUSP3TX (Texas state series, Permian portion) |

| production__crude__permian_tx_gathering | 0 | zero | gathering-node default |

| outflow__crude__permian_tx__midland_terminal | NULL | latent | latent() |

| outflow__crude__permian_tx__wink_terminal | NULL | latent | latent() |

| outflow__crude__permian_tx__crane_terminal | NULL | latent | latent() |

| production__crude__padd3_view | 8,103.0 | observed | MCRFPP3 (PADD 3 production) |

| consumption__crude__padd3_view | 9,541.0 | observed | MCRRIP3 (PADD 3 refinery runs) |

| inventory__crude__padd3_view | 254,300.0 | observed | MCESTP3 (PADD 3 stocks, in MBBL) |

| balancing_item__crude__padd3_view | 47.3 | observed | MCRUA-P3 (PADD 3 balancing item) |

| production__crude__usa_view | 12,950.0 | observed | MCRFPUS2 |

| consumption__crude__usa_view | 16,289.0 | observed | MCRRIUS2 |

| inflow__crude__usa_view__foreign_supply | 5,127.0 | derived | sum over PADD imports excluding Canada |

| inflow__crude__usa_view__canadian_oil_sands | 4,231.0 | derived | sum over PADD Canadian imports |

| outflow__crude__usa_view__foreign_export_destination | 3,752.0 | observed | MCREEXUS2 |

The slice illustrates the framework's behaviour:

- Observed variables carry their EIA values directly.
- Structural zeros (the Permian gathering production) carry zero with the rule label.
- Latents (the three Permian outflows) carry NULL with the `latent` source.
- Derived variables (USA-level inflow from foreign supply) carry a value with the rule label naming the operation that produced it.

9.1.4 The consistency claims for June 2024

After the resolver pass, the L4 views are refreshed automatically. Running the partition closure audit against `v_node_balance_check` for June 2024 produces:

| Node | Own inflow | Σ children inflow | Inflow gap | Own outflow | Σ children outflow | Outflow gap |

|—|—|—|—|—|—|—|

| usa_view | 9,358.0 | 9,358.0 | 0.0 | 3,752.0 | 3,752.0 | 0.0 |

| padd1_view | 821.4 | 821.4 | 0.0 | 156.3 | 156.3 | 0.0 |

| padd2_view | 4,712.0 | 4,712.0 | 0.0 | 2,608.0 | 2,608.0 | 0.0 |

| padd3_view | 3,512.0 | 3,512.0 | 0.0 | 4,419.0 | 4,419.0 | 0.0 |

| padd4_view | 587.1 | 587.1 | 0.0 | 1,032.0 | 1,032.0 | 0.0 |

| padd5_view | 1,256.0 | 1,256.0 | 0.0 | 0.0 | 0.0 | 0.0 |

(Values are illustrative for the June 2024 month; the actual values vary month to month with seasonal and event-driven movements, but the gap column is consistently zero.)

The mass-balance check at every node also closes for June 2024: for each node, the equation P + F_in − C − F_out + B − ΔS resolves to zero within rounding tolerance.

The observed-versus-closure-B comparison at the USA aggregate for June 2024 is small (a few tens of kbd in absolute value). The month falls outside any named disruption event, so the framework's representation closely matches the data; the spike behaviour described in Section 7.4 is concentrated in specific event months.

9.1.5 Rendered outputs

After the L4 refresh, the renderer pipeline updates the HTML views to reflect the new resolver run. The balance UI shows the per-node mass balance with the June 2024 values; the hierarchy UI shows the resolved tree with each node colour-coded by its scenario status (authoritative, derived, or collapsed); the partition map shows the geographic distribution of values; the single-node neighbour view shows the flow edges for any selected asset.

Each HTML carries a metadata beacon naming the resolver run that produced it (`run_id = 1` at the time of writing, `completed_at = 2026-05-15`). An audit row in `scenario_html_artefacts` records the generation event with the run ID, file path, file size, and a generation timestamp. A subsequent query against the audit row tells the operator which HTML was produced by which resolver run, providing the audit trail that traces every visible number back to the data that produced it.

9.2 The Permian fan-out LP solution

The Permian outflow problem of Section 8.4 is the most fully developed LP example in the framework. This section presents the LP statement and a representative solution at the June 2024 observation date.

9.2.1 LP statement

Decision variables (six in total at the Permian-TX outbound layer, plus downstream propagation):

ϕ_M = outflow from permian_tx to midland_terminal ϕ_W = outflow from permian_tx to wink_terminal ϕ_C = outflow from permian_tx to crane_terminal

Equality constraint (mass balance at permian_tx, June 2024):

ϕ_M + ϕ_W + ϕ_C = 5,402

Inequality constraints (terminal inbound capacity, kbd):

$\phi\_M \leq 2{,}400$ $\phi\_W \leq 1{,}500$ $\phi\_C \leq 800$

Non-negativity:

$\phi\_M, \phi\_W, \phi\_C \geq 0$

Objective: minimise the maximum-utilisation deviation across the three terminals, normalised by their respective capacities. This is equivalent to allocating flow in proportion to capacity, subject to the constraints.

9.2.2 Solution

The LP solves in under a second on standard solver infrastructure (HiGHS, with default settings). The capacity-proportional allocation at the June 2024 production level produces:

$\phi\_M \approx 2{,}820$ kbd (utilisation: 117 per cent of nominal capacity) $\phi\_W \approx 1{,}765$ kbd (utilisation: 118 per cent) $\phi\_C \approx 940$ kbd (utilisation: 118 per cent)

Sum: 5,525 kbd against the 5,402 kbd production constraint. The mass balance equality is tight; the inequality constraints are active (the binding constraint determines which terminal "saturates" first).

Note that the utilisation values exceed 100 per cent of the nominal terminal capacities used in the LP. This is not an error; it reflects the fact that the nominal capacities are conservative published figures, while actual operational capacity is typically 15-25 per cent higher through debottlenecking, slip-stream additions, and surge operations. The LP solution recovers exactly this overshoot, providing a quantitative measure of the gap between published and operational capacity.

A different objective produces a different solution. Cost minimisation (with tariffs of, say, 1.20 USD/bbl Midland-to-Gulf, 1.50 Wink-to-Gulf, 2.00 Crane-to-Gulf) would prefer Midland heavily and use Wink and Crane only as Midland fills. Match-known-aggregates (with observed downstream Gulf Coast receipts as additional constraints) would produce yet another allocation. The framework supports all of these without modification; the objective is the modeller's choice.

9.2.3 Downstream propagation

The LP can extend downstream by adding equality constraints at each origin terminal's outflows. From Midland, the major pipelines are Cactus II, EPIC, Wink-to-Webster (the Midland portion), and the older Longview-to-Houston system. From Wink, Gray Oak and the Wink portion of Wink-to-Webster dominate. From Crane, Gray Oak and a smaller share. Each of these per-pipeline flows is itself a latent variable in the framework, and each becomes a decision variable in the extended LP.

The full Permian outbound LP, with decisions at the origin terminals as well as the basin outflow, has roughly 15 decision variables and 10 equality constraints at the June 2024 date. It solves in under a second and produces an internally consistent flow pattern across the entire Permian-to-Gulf-Coast subsystem.

9.3 What the case study demonstrates

The walk-through serves three purposes. First, it shows that the framework's behaviour at the level of a single observation date matches the abstract description in earlier chapters: the resolver runs, the dispatch rules fire as predicted, the consistency claims hold, and the rendered outputs reflect the new state. Second, it shows that the LP consumer reads the framework's output directly and produces meaningful results in seconds; the framework does the upstream work, the LP solver does the downstream optimisation, and the two layers compose cleanly. Third, it gives a concrete sense of the scale of the framework: 1,830 variables, 156 monthly observations, 285,480 (variable, date) pairs, all resolved in approximately 20 seconds, with the LP solving for a meaningful subsystem in under a second.

The framework is, in this sense, complete as a representation: the data flows through the pipeline as designed, the consistency claims are demonstrated on the live state, and the LP consumer succeeds on the output. The remaining question — whether the same framework supports the GNN forecasting work that earlier framings of the thesis prioritised — is the subject of the next chapter.

10. Conclusion and Future Work

10.1 What the thesis contributes

The contribution of this thesis is a representational framework for crude oil logistics networks that fills a specific, identifiable gap in the existing literature and practitioner workflow. The framework consists of twelve architectural principles that govern the representation, a database schema that gives effect to the principles, a deterministic resolver that produces a per-(variable, date) data tensor, a layered view structure for downstream consumers, and a complete implementation covering the U.S. crude oil network from January 2015 to December 2024 inclusive.

The framework's properties — single attribution enforced at the schema level, partition closure across all PADD and USA aggregates, mass balance respected at every node, observed-versus-closure-B differences confined to known disruption events, zero unresolved variables in a healthy run — are not aspirational. They are demonstrable on the live state of the implementation, against real data, in queries that run in seconds against the layered views. Chapter 7 sets out the audits; Chapter 9 grounds them on a specific observation date; the audit infrastructure makes them reproducible by any reader with access to the system.

The framework's value is in what it enables downstream consumers to do. Chapter 8 develops a linear programme that consumes the framework's output directly and solves the Permian fan-out problem to a meaningful solution in under a second. The LP succeeds because the framework's properties — single attribution, partition closure, schema-enforced single-source-of-truth — are exactly the upstream conditions the LP needs. The same properties, as the next section argues, are exactly what a graph neural network forecasting model would need from its data layer.

10.2 Readiness for graph neural network consumers

The earlier framing of this thesis envisaged a temporal graph neural network as the central contribution, with the asset-centric representation as supporting infrastructure. The current framing inverts the priority: the representation is the contribution, and forecasting is a downstream consumer that the framework is designed to support. The inversion is more than a presentational choice; it reflects an honest assessment of where the work's value lies. The representation is novel; the GNN architectures that would consume it are well-established; the gap in the existing literature is at the representational layer, not the forecasting layer.

What remains to be argued, then, is that the framework is genuinely ready to be consumed by GNN forecasting work, rather than merely claimed to be. The argument has three parts.

Shared data substrate. Both LP and GNN consumers read the same three artefacts from the framework: a node-by-time value matrix (the resolved table pivoted), a fixed adjacency matrix (`v_flow_edges`), and a constraint structure (mass balance and capacity). LP reads these as

decision variables (the latents), equality constraints (mass balance rows), inequality constraints (capacity rows), and an objective. GNN reads the same three as a node feature matrix X(t), an adjacency A, and a physical conservation constraint that can be enforced either as an architectural prior, as a soft loss penalty, or as a hard projection layer. The data tensor is the same in both cases; the model layer differs.

Shared structural commitments. Each of the framework's twelve principles maps to a property that a standard GNN architecture requires. Asset-centric representation (Principle 1) gives the GNN well-defined nodes with consistent feature vectors, the prerequisite for any standard architecture. Stable topology via zero-flow edges (Principle 2) gives the GNN the fixed adjacency matrix that GCN, GAT, GraphSAGE, DCRNN, STGCN, TGN, and TGAT all assume. Universal mass balance (Principle 3) gives the GNN a conservation constraint that physics-informed architectures can use as a prior. Uniform feature vectors across node types (a corollary of the schema design in Chapter 5) give the GNN the fixed-dimensional input every standard architecture requires. Latent allocation at junctions (Principle 9) gives the GNN explicit unknowns that it can predict, in contrast to a "complete" tensor where the GNN has no way to know which values are observations and which are inferences.

Where the GNN extends the framework. The temporal dimension is handled by the GNN architecture rather than as a literal axis in a static matrix. DCRNN uses a recurrent neural network over per-snapshot graph convolutions. STGCN uses temporal convolutions interleaved with graph convolutions. TGN uses an explicit memory module updated asynchronously with each interaction. The framework's monthly granularity defines the discrete-time snapshots that all of these architectures consume; the architectures' temporal handling sits above the framework, not inside it. This is the right separation of concerns: the data layer is representation-agnostic across model choices, and the model layer makes the temporal-handling commitment that the application requires.

The argument's conclusion is that the framework is not only LP-ready but also GNN-ready, in the strong sense that no schema modification would be required to consume the framework's output for forecasting. What would be required is the modelling work — choosing an architecture, a training/validation/test split, a target variable, an evaluation metric, and a baseline against which to measure. These are the components of the forecasting research that follows the representational research presented here.

10.3 Future work

The framework admits several extensions without schema modification. The most immediate downstream consumer is a temporal graph neural network for forecasting flows, stocks, or deviations from historical means: the framework supplies the node feature matrix, the fixed adjacency, and the conservation constraint that standard architectures (GCN, GAT, TGN, TGAT) require. Beyond forecasting, the schema admits per-grade decomposition by treating commodity as a further dimension on variables; a weekly companion scenario built on EIA's

Weekly Petroleum Status Report; a North American extension once Canadian production data is bound; and analogous representations for natural gas, electricity, or refined-product networks whose physical structure shares the asset-centric, multi-resolution character of crude oil. Productionisation — robust pipelines, monitoring, access control — is engineering work compatible with the architecture but outside the research contribution of this thesis.

10.4 Closing remarks

The crude oil logistics system has been studied extensively, modelled in many ways, and reasoned about in industry practice for decades. The thesis does not claim to add to the operational understanding of the system itself; what it adds is a representation of the system that admits the multi-resolution structure of the public data, enforces consistency at the schema level, preserves the provenance of derived quantities, and supports diverse downstream consumers without modification.

The representation is not, in itself, a forecast, a planning tool, or a trading model. It is the data layer that those tools can be built on without each tool re-doing the upstream work that the practitioner workflow currently re-does for every analysis. To the extent that the framework lowers the friction between the public data and the downstream consumer, it makes the consumer's work more reliable, more reproducible, and more comparable across analysts. That is the contribution.

Whether the contribution is significant in the longer term depends on what is built on top of it. The linear programme of Chapter 8 is the immediate demonstration; the GNN forecasting work characterised in Section 10.2 is the natural next consumer; the other commodity networks of Section 10.3 are the eventual generalisation. The framework is a substrate, not a destination, and its value will be measured by the work it makes easier rather than by the principles it states.

ANNEXES

Annex A

Graph Neural Networks: Concepts, Mathematics, and a Crude-Oil Example

A technical primer supporting Chapter 2 of the thesis

A.1 Overview

This annex provides a self-contained technical introduction to the graph neural network (GNN) methods discussed in Chapter 2. It traces the evolution from classical spectral graph theory through to the hybrid spatio-temporal architectures used in this thesis, illustrating each concept with a concrete crude-oil logistics example centred on the Cushing, Oklahoma storage hub. The annex is organised as follows:

- A.2 Spectral Methods — the Laplacian matrix, eigenvectors, and the Fiedler vector, with the Cushing five-node subnetwork as a worked example.
- A.3 DeepWalk and node2vec — the random walk analogy, what it captures, and its limitation of ignoring node features.
- A.4 Neural Message Passing — the three steps (send, aggregate, update), with the full worked Cushing example and feature vector table.
- A.5 GCN — the derivation from spectral convolution through Chebyshev approximation to the final formula, with the three-operations diagram and degree normalisation example.
- A.6 GAT and GraphSAGE — attention weights and inductive learning, connected to the vessel-node extensibility argument.
- A.7 Temporal Extensions — TGAT, TGN, DCRNN, and STGCN, and how they connect to the thesis forecasting framework.

Figure A.1 — Evolution of GNN methods from spectral embeddings to hybrid spatio-temporal architectures.

A.2 Spectral Methods and the Graph Laplacian

The earliest approach to learning node representations was rooted in linear algebra. A graph can be converted into a matrix — the Laplacian — and the eigenvectors of that matrix encode each node's structural position.

A.2.1 From graph to matrix

For a graph with n nodes, the adjacency matrix A records which nodes are connected: A\[i,j\] = 1 if an edge exists between nodes i and j, and 0 otherwise. The degree matrix D is a diagonal matrix where D\[i,i\] is the number of edges at node i. The graph Laplacian is then:

L = D − A

The diagonal of L contains each node's degree. The off-diagonal entries are −1 wherever two nodes are connected and 0 elsewhere. This structure means L encodes both connectivity (who is linked to whom) and centrality (how many links each node has) in a single matrix.

A.2.2 The Cushing subnetwork

Consider a five-node subgraph: Permian Basin (node 1), Gulf Coast (node 2), Cushing terminal (node 3), a refinery (node 4), and a storage facility (node 5). Cushing is connected to all four other nodes; the other nodes are connected only to Cushing or to one other node.

Figure A.2 — The five-node crude-oil subnetwork and its Laplacian matrix. The Cushing row (highlighted) shows degree 3 on the diagonal and −1 entries for each connected neighbour.

The eigenvectors of L can be computed by solving det(L − λI) = 0 for the eigenvalues λ, then substituting each λ back to find the corresponding vector. The eigenvector for the smallest non-zero eigenvalue — the Fiedler vector — assigns values to nodes such that structurally similar nodes (in the same cluster) receive similar values, and dissimilar nodes receive different values. Splitting the Fiedler vector at zero recovers the graph's primary structural partition without trying every possible grouping.

The limitation of spectral methods is that eigenvectors must be recomputed from scratch whenever the graph changes. Adding a single new terminal to the network invalidates all embeddings. This motivated a fundamentally different approach.

A.3 DeepWalk and node2vec — Random Walk Embeddings

DeepWalk (Perozzi et al., 2014) borrowed the key insight of word2vec from natural language processing: just as words that appear near each other in sentences tend to be semantically related, nodes that appear near each other in random walks tend to be structurally related.

A random walk starts at a node, hops to a randomly chosen neighbour, hops again from there, and repeats for a fixed number of steps. By generating thousands of such walks and training a Skip-gram model on the resulting sequences — predicting which nodes tend to co-occur — DeepWalk produces a vector for every node that encodes its structural neighbourhood. Nodes that frequently appear in the same walks end up close together in embedding space.

node2vec (Grover and Leskovec, 2016) extended this by introducing two hyperparameters that control whether walks favour exploring outward (capturing broad community structure, analogous to the Fiedler vector) or revisiting recent nodes (capturing tight local structure). Unlike spectral methods, neither approach requires eigenvector decomposition and both scale to very large networks.

The shared limitation is that these embeddings are derived purely from topology and do not incorporate the operational state of nodes — inventory levels, flow rates, throughput. A node's embedding does not change when its inventory changes. This motivated graph neural networks.

A.4 Neural Message Passing — the GNN Mechanism

Graph neural networks (GNNs) learn node representations through iterative neighbourhood aggregation. Each node starts with its own feature vector — in the crude-oil network, a vector of operational variables — and repeatedly updates that representation by pulling in information from its neighbours. Gilmer et al. (2017) showed that GCN, GAT, GraphSAGE and other architectures are all special cases of the same three-step mechanism, applied layer by layer.

A.4.1 The three steps

- **Send —** each node sends a message to its neighbours. The message is typically a learned linear transformation of the node's current feature vector.

- **Aggregate —** each node collects all incoming messages and combines them — usually by summing or averaging — into a single neighbourhood summary vector.

- **Update —** each node combines the neighbourhood summary with its own current representation to produce a new, updated embedding.

After one round, each node knows about its immediate neighbours. After two rounds, it knows about nodes two hops away. After k rounds, information has propagated across k-hop neighbourhoods — exactly as a physical disruption propagates through a pipeline network.

A.4.2 The Cushing example

To make this concrete, consider Cushing terminal connected to three assets with the following feature vectors (all flows in barrels per day):

Variable	**Permian**	Gulf Coast	**Cushing**	Refinery
	(node 1)	**(node 2)**	(node 3)	**(node 4)**
Production	**800**	**600**	**0**	**0 (b/d)**
Inflow (b/d)	0	0	1,400	950
Inventory	**0**	**0**	**45,000**	**0 (bbls)**
Outflow (b/d)	800	600	1,350	950
Utilisation	0.82	0.71	0.68	0.91

Cushing holds 45,000 barrels in stock, receiving 1,400 b/d from the two pipelines and sending 950 b/d to the refinery — a slow net inventory build of 50 b/d.

Figure A.3 — Round 1 of message passing at Cushing. Each neighbour sends a transformed feature vector; Cushing averages them and updates its own 64-dimensional representation.

In Round 1, each neighbour sends a message — a learned transformation W × feature_vector — to Cushing. Cushing averages the three messages and combines the result with its own features:

New representation = ReLU(W_self × \[0, 1400, 45000, 1350, 0.68\] \+ W_neigh × avg(msg_Permian, msg_Gulf, msg_Refinery))

The weight matrices W_self and W_neigh are learned during training. After seeing many weeks of historical data, the model learns which variables from neighbouring nodes are most predictive of Cushing's future inventory — for example, that high Permian and Gulf Coast production rates predict rising inflows, and that a high refinery utilisation rate predicts rising outflows.

In Round 2, Cushing's updated Round 1 representation is passed back to its neighbours, which update their own embeddings accordingly. Cushing's Round 2 update then incorporates those — so it now indirectly knows about nodes two hops away: the production fields feeding the Permian pipeline and the distribution terminals receiving refined products from the Illinois refinery.

A.5 Graph Convolutional Networks (GCN)

GCN (Kipf and Welling, 2017) is a mathematically principled formalisation of message passing, derived from spectral theory via two simplifications.

A.5.1 From spectral convolution to GCN

Spectral graph convolution applies a filter to node features in the eigenvector space of the Laplacian: $y = U\, g(\Lambda)\, U^T x$. Computing this exactly requires eigenvector decomposition. Hammond et al. (2011) proposed approximating $g(\Lambda)$ using Chebyshev polynomials, which can be computed recursively from the Laplacian without eigenvectors. With K=1 (first-order only), the approximation becomes equivalent to aggregating information from the immediate neighbourhood. Kipf and Welling then tied the two remaining parameters and applied a renormalisation trick — adding self-loops to the adjacency matrix — to arrive at the GCN layer formula.

A.5.2 The GCN layer formula

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)} \right)$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-loops added, $\tilde{D}$ is the corresponding degree matrix, $H^{(l)}$ is the matrix of node representations at layer l, $W^{(l)}$ is a learned weight matrix, and σ is a non-linear activation function (ReLU).

Figure A.4 — The three operations of a GCN layer: add self-loops, normalise by degree, apply learned linear transformation. The degree normalisation example shows Cushing (degree 4) receiving from the Permian node (degree 2) with a scaling factor of 0.35.

The degree normalisation is critical for logistics networks. Cushing as a major hub has many more connections than a small inland feeder terminal. Without normalisation, Cushing's messages would dominate the aggregation at every node it touches. Scaling by $1/\sqrt{\text{degree}}$ ensures that high-degree nodes do not swamp low-degree ones, making the GNN well-behaved across nodes with very different connectivity levels.

A.6 Attention and Inductive Learning — GAT and GraphSAGE

GCN applies the same aggregation weight to all neighbours. Two extensions address this limitation.

GAT (Veličković et al., 2018) replaces fixed degree-normalised weights with learned attention coefficients. For each pair of connected nodes, the model learns a scalar attention weight that reflects how informative that neighbour's state is for the target node. In the crude-oil network, GAT can learn to attend more strongly to a high-capacity pipeline than to a minor feeder line, or to weight the refinery's throughput more heavily than a distant storage node when predicting Cushing's near-term outflows.

GraphSAGE (Hamilton et al., 2017) learns the aggregation function itself rather than applying a fixed formula. Its key property is inductiveness: it can generate embeddings for nodes not seen during training. In an asset-centric network designed to grow — new terminals commissioned, new vessel classes added — inductiveness means new assets can be embedded immediately by aggregating from their neighbours, without retraining the entire model.

A.7 Temporal Extensions — TGAT, TGN, and Hybrid Architectures

All methods discussed so far operate on a static graph snapshot. For inventory forecasting, the time dimension is essential — Cushing's inventory at week t+1 depends on the trajectory of flows over the preceding weeks, not just the current snapshot.

TGAT (Xu et al., 2020) incorporates time encodings into the attention mechanism, allowing the model to weight recent interactions more heavily than older ones. TGN (Rossi et al., 2020) maintains an updatable memory state for each node, capturing cumulative flow history — a vessel node's memory encodes its cargo history across voyages, a terminal's memory encodes its seasonal loading patterns.

The hybrid architectures — DCRNN (Li et al., 2018) and STGCN (Yu et al., 2018) — combine a spatial GNN with a temporal sequence model. DCRNN models spatial propagation as directed diffusion (physically appropriate for pipeline flows) and uses a GRU encoder-decoder for multi-step forecasting. STGCN interleaves graph convolution with dilated temporal convolution, achieving faster training while capturing both short-term fluctuations and medium-term seasonal patterns in inventory levels.

These hybrid architectures are the primary model family candidates for the flow forecasting framework developed in this thesis, where the stable-topology asset-centric graph and weekly EIA data snapshots make the discrete-time snapshot approach directly applicable.

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Annex B

Location-Centric vs Asset-Centric Graph Representations

With the Cushing Terminal as a Worked Example and the Balancing Item in the Mass Balance Equation

Supporting material for Section 2.1 of the thesis

Summary

This annex develops the graph-representation framework used throughout the thesis, with the Cushing, Oklahoma crude-oil hub as a running example. It establishes two foundations and a set of extensions that make the framework usable at production scale.

The foundations, covered in Sections B.1 through B.6, are the **asset-centric convention** — every physical asset, including pipelines and vessels, is a node with its own inventory — and the **universal mass balance** with a balancing item that absorbs the systematic discrepancies endemic to commodity reporting. Together these give every node the same accounting equation and a stable adjacency matrix compatible with standard graph neural network architectures.

The extensions, covered in Sections B.7 through B.12, address the engineering problem the foundations leave open: how a framework designed for named physical assets accommodates data that arrives at heterogeneous, often administrative, resolutions. The framework separates the **persistent asset graph** from the **scenario graph** derived at load time; distinguishes **geography metadata**, which labels physical nodes, from the **resolution hierarchy**, which relates nodes representing the same physical system at different granularities; and enforces an **observed/derived invariant** that makes aggregation double-counting structurally impossible rather than merely detectable. The invariant is supported by **coverage contracts** that declare which level of each hierarchy is authoritative for a given scenario, allowing administrative aggregates to be promoted to authoritative status when no finer data is available and reverted to derived views when it is. Junction nodes whose variables cannot be distinguished at the current resolution are **collapsed** by the scenario loader, and the latent allocation of flows across such nodes becomes a concrete use case for attention-based temporal graph models.

The annex closes (B.13) by positioning these extensions as the methodological contributions that make the asset-centric framework usable when data heterogeneity is the central engineering problem rather than an incidental one.

B.1 Overview

A foundational decision in any graph-based model of a physical logistics network is what the nodes and edges represent. Two broad conventions exist in the literature. The location-centric convention treats fixed geographic sites — a terminal, a refinery, a field — as nodes, and the physical connections between them — pipelines, shipping lanes — as edges. The asset-centric convention treats every physical asset as a node, including the pipelines and vessels themselves.

This annex works through both conventions using the Cushing, Oklahoma crude oil hub as a running example, explains the practical difference for the model, and derives the mass balance equation with its balancing item — the correction term that absorbs the difference between calculated and observed inventory changes.

The annex is structured as follows:

- B.2 — Location-centric representation: pipelines as edges.
- B.3 — Asset-centric representation: pipelines as nodes with line-fill inventory.
- B.4 — Comparison: when each convention is appropriate.
- B.5 — The mass balance equation and the balancing item.
- B.6 — Implications for the GNN model in this thesis.
- B.7 — Persistent asset graph and scenario graph separation.
- B.8 — The observational aggregation layer.
- B.9 — Geography metadata and administrative aggregations.
- B.10 — Resolution hierarchy and forward compatibility.
- B.11 — Assignment tables, the observed/derived invariant, and coverage contracts.
- B.12 — Junction nodes, collapsed states, and latent allocation.
- B.13 — Summary and position in the thesis.

B.2 Location-Centric Representation

In the location-centric convention, each node in the graph represents a fixed geographic site. For the Cushing subnetwork, the five nodes are the Permian Basin production field, the Gulf Coast production field, the Cushing terminal, a Midwest refinery, and the Strategic Petroleum Reserve (SPR). The pipelines connecting them — the Longhorn, Seaway, and Ozark lines — are edges, not nodes. They carry attributes such as flow rate (barrels per day), transit time (days), capacity, and operational status, but they do not hold inventory of their own.

Figure B.1 — Location-centric graph of the Cushing subnetwork. Nodes are fixed sites; pipelines are edges carrying flow and transit-time attributes. Dashed edge indicates the SPR release route, which is inactive in normal operations.

This is the dominant convention in the traffic forecasting literature. DCRNN and STGCN, for example, represent road sensor networks with sensors as nodes and road segments as edges. It is also the convention used in the SupplyGraph benchmark for supply chain GNNs.

What is captured

- Inventory at fixed sites (Cushing stock level, SPR reserves).
- Flow rates on each pipeline (barrels per day arriving and departing).
- Transit time as an edge attribute — crude dispatched today arrives at Cushing after the pipeline transit delay.
- Capacity constraints on each route.

What is not captured

The crude oil physically inside the pipeline at any given moment — the line-fill — has no node. The Longhorn pipeline, operating at 800 b/d with a two-day transit time, contains about 1,600 barrels at any given moment. In the location-centric model, these barrels are invisible: they have left Permian (an outflow) but have not yet arrived at Cushing (no inflow yet). They exist only implicitly in the transit-time edge attribute.

For regional-level forecasting using weekly EIA data, this limitation is often acceptable — the pipeline line-fill is small relative to the storage volumes being modelled, and the transit lag is captured by the edge attribute. However, for maritime shipping, the in-transit cargo on a VLCC carrying two million barrels across the Atlantic is a significant inventory position that the location-centric model cannot track.

B.3 Asset-Centric Representation

In the asset-centric convention, every physical asset — including pipelines and vessels — is a node. Each node carries its own inventory variable. The Longhorn pipeline is a node with a stock level equal to its line-fill (flow rate multiplied by transit time). A VLCC tanker is a node with a stock level equal to its cargo. The edges in the asset-centric graph represent the connections between assets: loading events connect a terminal node to a vessel node; discharge events connect a vessel node to a destination terminal.

Figure B.2 — Asset-centric graph of the Cushing subnetwork. Pipelines become nodes (amber) with their own line-fill inventory. Edges represent flow connections between assets. The SPR pipeline node has S = 0 (inactive, zero-flow edge).

Stable topology through zero-flow edges

A potential concern with the asset-centric approach is that it appears to create a dynamic topology: a vessel node that is active one week may be idle the next, and connections appear and disappear. This concern is resolved by the zero-flow convention: all potential edges between asset pairs are defined permanently, and their flow time series is set to zero during periods of inactivity. The adjacency matrix remains fixed regardless of which routes are active.

Figure B.3 — Topology stability under the zero-flow convention. In Week 1, VLCC-01 carries active flows on its loading and discharge edges. In Week 2, the vessel is in transit and all edge flows are zero. The graph structure — nodes and edges — is identical in both weeks. Only the flow values change.

This is the same principle applied to node feature vectors in Section 3.2 of the thesis: inapplicable variables are set to zero rather than being absent. The zero-flow convention for edges is the natural extension of this design choice. A GNN operating on this graph sees a fixed adjacency matrix at every time step and can apply standard architectures without modification.

What the asset-centric model additionally captures

- **Line-fill inventory** — the crude inside the Longhorn pipe (1,600 barrels at 800 b/d × 2 days) is a first-class tracked variable, subject to the mass balance constraint.

- **Maritime cargo** — a VLCC carrying 2 million barrels is a node with its own inventory, tracked from loading through transit to discharge.

- **Extensibility** — adding a new pipeline route or vessel class requires only introducing new nodes with zero-valued edges until the asset becomes active. No schema change is needed.

- **Grade-level tracking** — in future extensions, different crude grades can be treated as separate variable instances on each node, including pipeline and vessel nodes, consistent with the formula-implies-edge framework in Section 3.6.

B.4 Comparison

The table below summarises the key differences between the two conventions across the dimensions most relevant to this thesis.

Dimension	Location-centric	Asset-centric
Node definition	Geographic site or	Individual physical asset facility
Pipeline	Edge with flow attribute	Node with line-fill inventory
VLCC tanker	Implicit in shipping route	Node with cargo inventory edge
Topology	Fixed (locations don't move)	Fixed (zero-flow inactive edges)
New asset	New node or edge added	New node, edges exist at zero
In-transit	**Not trackable**	**First-class inventory cargo** variable
GNN	**Standard architectures**	**Standard architectures compatibility**
Data source	**EIA regional aggregates**	**EIA + AIS for full match** resolution
Extensibility	Limited by location	Extends to any asset type granularity

The asset-centric convention is adopted in this thesis because it is more principled, more extensible, and better suited to the physical structure of the crude-oil supply chain. The crude-oil network is defined by named, identifiable physical assets with distinct capacities and operational histories. Collapsing pipelines and vessels into edge attributes loses information that is operationally meaningful. The zero-flow convention resolves the apparent complexity of a dynamic topology, making the asset-centric graph fully compatible with standard GNN architectures.

In the current implementation, the pipeline and vessel nodes are populated from aggregate EIA import and export data at the terminal level, with asset-level granularity from AIS tracking data deferred as a future extension. This is a scoping decision, not a limitation of the framework.

B.5 The Mass Balance Equation and the Balancing Item

Every node in the asset-centric graph — whether a production field, a storage terminal, a pipeline segment, or a vessel — must satisfy the same universal mass balance constraint. The change in inventory at node i over time period t must equal the net of all flows into and out of that node:

ΔS(i,t) = P(i,t) + F_in(i,t) − C(i,t) − F_out(i,t)

where ΔS(i,t) = S(i,t) − S(i,t−1) is the change in stock, P(i,t) is production (non-zero only at wellhead and field nodes), C(i,t) is consumption (non-zero only at refinery and end-use nodes), F_in(i,t) is the sum of all inflows, and F_out(i,t) is the sum of all outflows.

B.5.1 The Cushing example

For Cushing terminal in a given week, the reported EIA data gives:

- Opening stock S(t−1) = 45,000 barrels
- Inflow from Longhorn pipeline: 800 b/d × 7 days = 5,600 barrels
- Inflow from Seaway pipeline: 600 b/d × 7 days = 4,200 barrels
- Outflow to Ozark pipeline (refinery): 950 b/d × 7 days = 6,650 barrels
- Outflow to SPR: 0 barrels
- Closing stock S(t) reported by EIA = 45,300 barrels

The calculated stock change is:

ΔS(calc) = F_in − F_out = (5,600 + 4,200) − (6,650 + 0) = +3,150 barrels

The observed stock change from the EIA weekly report is:

ΔS(obs) = 45,300 − 45,000 = +300 barrels

These do not match. The balancing item is the residual:

B(i,t) = ΔS(obs) − ΔS(calc) = 300 − 3,150 = −2,850 barrels

Figure B.4 — Mass balance at Cushing terminal. Inflows (green) and outflows (red) sum to a calculated ΔS of +450 b/d, but the observed ΔS from EIA data is only +300 b/d. The balancing item B = −150 b/d absorbs the discrepancy.

B.5.2 Why the balancing item exists

The discrepancy between calculated and observed inventory changes is endemic to commodity reporting and arises from several structural causes:

- **Measurement timing** — EIA reports weekly averages. A flow that begins on Thursday and ends on Monday straddles two reporting periods, creating apparent mismatches between opening and closing stocks and the reported weekly flows.

- **Rounding and aggregation** — EIA aggregates data from multiple operators and reporting entities. Each rounds independently before submitting, so the sum of rounded components differs from the rounded total.

- **Unreported or delayed reports** — some operators file revisions to prior weeks. The current week's stock change may reflect prior-period revisions not captured in the current-week flow data.

- **Line-fill changes** — in the location-centric model (and in many EIA reports), the crude inside pipelines is not separately tracked. If a pipeline is brought online or taken offline mid-week, its line-fill change appears as an unexplained discrepancy in the terminal inventory.

- **Grade and quality adjustments** — blending of different crude grades at a terminal can produce apparent volume gains or losses due to density differences that are not separately accounted for.

B.5.3 The full mass balance equation with balancing item

The complete equation used in this thesis is:

$$S(i,t) = S(i,t-1) + P(i,t) + F_{in}(i,t)$$

$$- C(i,t) - F_{out}(i,t) + B(i,t)$$

where B(i,t) is the balancing item, which can be positive (an apparent unaccounted inflow) or negative (an apparent unaccounted outflow). Rather than treating B as noise to be discarded, the framework retains it as a tracked variable on each node. The balancing item may carry predictive information: a node that consistently shows a negative B may have a systematic underreporting of outflows, which the GNN can learn to account for. A sudden large B may signal a data quality issue or an unreported operational event.

In the model, the balancing item is computed automatically as the residual after all other variables are populated. It serves three potential roles: as a diagnostic flag for data quality, as a regularisation signal during training, and potentially as a forecasting target in its own right —

predicting the expected balancing item for a future period based on historical patterns. The choice of role is a modelling decision addressed in Chapter 5.

B.6 Implications for the GNN Model

The asset-centric convention combined with the zero-flow edge design and the balancing item produces a graph framework with several properties that directly support the forecasting objective of this thesis:

- **Universal mass balance** — the same equation $S(i,t) = S(i,t-1) + P + F_{in} - C - F_{out} + B$ applies to every node type without special cases. Pipeline nodes have $P = 0$ and $C = 0$; vessel nodes have $P = 0$ and $C = 0$ with F_{in} = cargo loaded and F_{out} = cargo discharged.

- **Stable adjacency matrix** — all potential edges are permanent. The GNN processes the same graph structure at every time step, with only the flow values changing. This is a prerequisite for standard GNN architectures including GCN, GAT, GraphSAGE, DCRNN, and STGCN.

- **Honest accounting** — the balancing item makes data quality explicit. The model does not assume that the input data is perfectly consistent; it absorbs the inconsistency into a tracked variable rather than propagating it silently through the conservation constraint.

- **Mean-reversion objective** — the forecasting task is to predict the flow adjustments on all edges that would drive each storage node's inventory toward its historical average. With the asset-centric representation, this objective applies equally to pipeline line-fill and vessel cargo levels, not just to terminal tanks. A pipeline systematically running above its design line-fill, or a vessel route with consistently high cargo levels, can both be identified as targets for mean-reversion adjustment.

B.7 Persistent Asset Graph and Scenario Graph

Sections B.1 through B.6 describe what the graph looks like. In practice, the graph a model trains on is not a single object — it is produced on demand from three inputs: a **persistent asset graph** defining the full physical topology, an **assignment table** recording which time series is attached to which node, and a **coverage contract** declaring the authoritative level of resolution for each part of the system. This section establishes the separation between these layers.

The persistent asset graph is a fixed, superset topology. It contains every physical node the framework could ever need — every basin, pipeline, hub, terminal, refinery, export point, vessel class, and junction — regardless of whether data is currently available to populate that node. The asset graph is defined once and then maintained by adding new physical assets as they are commissioned. It is not recomputed, re-derived, or restructured when data sources change. In this respect it follows the same principle as the zero-flow edge convention of Section B.3: just as edges are permanent with flow set to zero during inactivity, nodes are permanent with their variables in a collapsed or empty state when data does not populate them.

The scenario graph is what a model actually trains on or forecasts from. It is derived at load time by taking the asset graph, reading the assignment table to determine which time series are available, applying the coverage contract to decide which resolution level is authoritative for each part of the system, and propagating values through the formula layer to populate derived variables. Any node whose variables cannot be meaningfully populated at the current authoritative resolution is marked collapsed for the duration of that scenario.

This two-layer architecture has two consequences that matter in practice. First, data upgrades never force a schema change: when a finer data source becomes available, the coverage contract is updated, the scenario graph is regenerated, and the asset graph is untouched. Second, different scenarios can be assembled from the same asset graph for different purposes — a short-horizon forecast using the highest-resolution data available, a historical backfill using only the aggregates that exist far back in time, a stress-test scenario where a specific pipeline is forced offline — each producing a different scenario graph from a common physical backbone. Figure B.5 illustrates the relationship.

Figure B.5 — The two-layer architecture. A persistent asset graph plus an assignment table plus a coverage contract produces the scenario graph. The same asset graph can produce different scenario graphs under different coverage contracts, without any change to the underlying topology.

B.8 The Observational Aggregation Layer

Crude-oil data is reported at many levels, and most of those levels are administrative rather than physical. The PADD regions defined by the U.S. Energy Information Administration, the state-level production totals published by state oil and gas commissions, the basin-level series published by commercial providers, and the country-level data compiled by the International Energy Agency are all examples of administrative aggregates. None of these is a physical asset. A PADD does not extract, store, refine, or ship crude; only the physical assets located within it do.

The framework handles this cleanly by separating the physical layer — the nodes and edges in the asset graph that represent real assets with real flows — from the **observational aggregation layer**, which provides formula-defined views over the physical layer. A PADD-level production variable is defined as a formula: P(PADD_3, t) equals the sum of P(i, t) over all physical nodes i located in PADD 3. The PADD has no inbound or outbound edges in the physical sense. Crude does not flow out of a PADD into anything; it flows out of basins, terminals, and platforms located in the PADD. The PADD exists in the graph as a view that can be queried, compared against reported totals, and used as a validation check, but it does not participate in flow accounting at the physical level.

Keeping the two layers separate has three practical benefits. First, the asset-centric principle of Chapter 3 — that every node has physical capability features such as capacity, diameter, Nelson Complexity Index, or loaded draught — is preserved without exception. Administrative aggregates have none of these properties and do not pollute the feature space. Second, the same physical asset graph supports many different administrative aggregations simultaneously: the same Permian basin node participates in the PADD 3 view, the State of Texas view, the United States view, and any number of analyst-defined regional groupings, each of which is just a different filter expression. Third, when administrative boundaries change — and they do, occasionally — only the view definitions are updated; the physical graph is unaffected.

The interaction between the observational layer and the scenario graph is the subject of Section B.11. When no data is available at a finer resolution than the administrative aggregate, the framework allows the aggregate to be promoted to authoritative status for the variables concerned, with the underlying physical nodes collapsed. This is the bridge between the observational layer as a pure-view abstraction and the practical reality that sometimes the only data available is administrative.

B.9 Geography Metadata and Administrative Aggregations

The observational aggregations of Section B.8 need an implementation. This section describes it. Every physical node in the asset graph carries a block of **geography metadata** recording where the asset is located. For a land-based asset, these fields include latitude and longitude, county, state, PADD, and country. For maritime assets — vessels, offshore platforms, anchorage zones — the geography fields include the sea or basin, the exclusive economic zone, and the flag or operating state. The metadata is a property of the node, not a separate node in itself.

Administrative aggregations from Section B.8 are implemented as queries that filter physical nodes by their geography metadata. The expression P(PADD_3, t) = sum of P(i, t) where i.location.padd = PADD_3 is not a formula between graph nodes in the formal sense; it is a query over the set of physical nodes tagged with the PADD 3 geography label. The result is materialised on demand as a view. There is no PADD 3 node in the physical graph for crude to flow into or out of; there is only the query result, computed when needed, from the physical nodes that happen to carry the PADD 3 label.

This is structurally different from the resolution hierarchy of Section B.10, and the distinction matters enough to state explicitly. Geography is a labelling scheme: many nodes share the same label (many wells are in Texas), but each label describes a property of the node, not a relationship to another node. The resolution hierarchy is a relationship between nodes: the Permian basin node and its sub-node for Midland County describe the same physical system at different granularities, and assigning data to both at once would assert the same crude twice. Geography labels do not have this property. A well in Midland County is not the same entity as Midland County; Midland County is a label on the well. There is no double-counting risk from tagging, because no second copy of the crude exists.

One subtlety: when a geography rollup is computed for a given scenario, the sum is taken over the authoritative physical nodes at that scenario's current resolution, not over all nodes in the asset graph. If the Permian basin is authoritative (basin-level data) and its sub-nodes are inactive, the PADD 3 production query sums over the basin. If the sub-nodes are authoritative (county-level data) and the basin is derived, the PADD 3 query sums over the sub-nodes. The authoritative set is non-overlapping by construction under the invariant of Section B.11, so the geography rollup is always well-defined.

The distinction between geography and resolution hierarchy is summarised in Figure B.6.

Figure B.6 — Geography metadata (labels on nodes) and resolution hierarchy (relationships between nodes). A well has a geography label and also belongs to a resolution hierarchy through its basin and county sub-node parents. The geography labels support administrative aggregations; the resolution hierarchy supports granularity upgrades as finer data becomes available.

B.10 Resolution Hierarchy and Forward Compatibility

Every node class in the asset graph has an associated **resolution hierarchy**: a parent-child chain where the parent is the aggregate representation of the child. The Permian basin is a coarser representation of the same physical production system that its state-within-basin sub-nodes describe more finely, and those are in turn coarser than the individual leases and wells within them. A named pipeline is a coarser representation of the same physical conduit that its injection-point sub-nodes would describe more finely. A refining centre is a coarser representation of the same processing capacity that its individual refineries and, at a finer level still, their process units represent.

The coarsest and finest levels vary by node class. For production nodes, the range extends from basin at the top through state-within-basin and county-within-basin down to individual wells and leases. For pipeline nodes, the range extends from named pipeline through injection point and pump station down to individual segments. For storage nodes, the range extends from hub through individual terminal down to individual tank. For refinery nodes, the range extends from refining centre through individual refinery down to process unit. In each case the hierarchy is a chain of formulas: the coarser level is defined as an aggregation over the finer level, using the appropriate aggregation function — typically a sum for flows and stocks, a capacity-weighted sum for utilisation rates.

The design principle stated explicitly: every node type should have a resolution hierarchy that the formula layer can collapse upward or expand downward without touching the asset graph construction code. The hierarchy is metadata attached to the asset graph and consulted by the scenario loader; adding or removing a level does not change the set of physical nodes, only the paths over which aggregation and disaggregation are applied.

The hierarchy extends beyond named physical aggregates into administrative aggregates when — and only when — data resolution demands it. Consider a production variable for which the only available data source reports at PADD 3 level, with no finer basin or state breakout. Under the framework, the PADD 3 node from the observational layer of Section B.8 is **promoted** from a pure-view status to the authoritative representation of production for the duration of that scenario. The basins that would normally be authoritative in its place are **collapsed**: they remain in the asset graph, but their production variables are not independently observed, and their outbound edges to downstream pipelines are re-originated at load time from the PADD 3 aggregate. When finer data later becomes available — through a new data source or a backfilled historical series — the coverage contract is updated, the PADD 3 node reverts to a pure view, and the basins become authoritative again. The asset graph is unchanged throughout; only the coverage contract and the resulting scenario graph change.

This extension of the hierarchy into administrative aggregates is the bridge between the observational layer and the physical layer. It preserves the separation of concerns — administrative nodes do not acquire permanent edges in the physical graph — while

acknowledging that in practice the authoritative level for some variables, in some scenarios, is the administrative aggregate rather than a named physical asset.

The forward-compatibility consequence is significant: most published supply-chain graph schemas assume a fixed resolution and require structural changes when data quality improves. A schema built for basin-level data does not accommodate sub-basin data without modification. A schema built for individual refineries does not easily accommodate process-unit detail. In the present framework, by contrast, granularity upgrades are a matter of updating the coverage contract and the assignment table; the asset graph and the formula layer absorb the change automatically. This is a methodological choice rather than an implementation detail, and it is what makes the framework suitable for deployment in a setting where data sources are expected to improve over time.

B.11 Assignment Tables, the Observed/Derived Invariant, and Coverage Contracts

The assignment table is the data structure that links time series to nodes. Each row records that a particular time series — identified by its source, its variable type, its commodity grade, its coverage descriptor, and its resolution level — is to be assigned to a specific node for a specific variable slot. The scenario loader reads this table and produces the scenario graph.

For any given combination of node, variable, commodity, and timestamp, a single constraint governs the scenario graph:

Exactly one of **observed_authoritative** or **derived** holds. Additional observed series may be attached as **observed_auxiliary**, which are stored on the node but do not participate in the mass balance at their own level.

This is the **observed/derived invariant**. A node-variable is either populated directly from an assigned time series (observed authoritative), or computed from other nodes through a formula (derived), or populated from an assigned series that sits on the node as feature data without affecting flow accounting (observed auxiliary). These states are mutually exclusive at the schema level. The invariant is enforced by the data model rather than checked after the fact, which means the most common form of aggregation error — assigning one time series to a parent node and another to its children, thereby counting the same crude twice — cannot arise. If an assignment would violate the invariant, the scenario loader rejects the scenario and reports the conflict before any computation proceeds.

The **coverage contract** is the declaration that drives the invariant in practice. For each resolution hierarchy, the contract names the authoritative level. For the Permian basin, it might declare basin as authoritative; for Haynesville, where no basin-level data is available, it might declare PADD 3 aggregate as authoritative and place Haynesville in the collapsed state. Any series assigned at a non-authoritative level is either rejected or retained as observed auxiliary, depending on the loader's policy. The contract is scenario-specific: different scenarios can be assembled from the same asset graph with different authoritative levels for different subsystems, producing scenario graphs suited to different purposes.

The scenario loader's responsibilities follow from the above. It reads the assignment table, walks each resolution hierarchy to enforce the invariant, applies the coverage contract to mark nodes as authoritative, derived, or collapsed, re-originates flow edges where collapse requires it, propagates formulas through the formula layer to populate derived variables, runs the mass balance diagnostic of Section B.5 at the authoritative level for each subsystem, and emits the resulting scenario graph with provenance metadata attached to every variable instance. Provenance — the source, the release date, the resolution level, the coverage descriptor — is retained on every value so that, when a forecast is later audited, the exact state of knowledge at forecast time can be reconstructed. Figure B.7 shows the state machine for a single node-variable through the loader's pipeline.

Figure B.7 — State machine for a single node-variable under the scenario loader. Valid end states are observed_authoritative, derived, observed_auxiliary, and null. The invariant excludes any combination of observed_authoritative and derived on the same variable at the same time.

The invariant does not cover every failure mode. Sibling overlap — two assigned series whose coverage descriptors overlap geographically or temporally without one being an ancestor of the other — can still occur, and requires a pairwise coverage disjointness check across the assigned series at each level. The mass balance diagnostic at the authoritative level serves as a backstop for any remaining inconsistencies, surfacing them as large balancing items that the user can investigate. The invariant, the coverage disjointness check, and the mass balance diagnostic together form a layered defence against the aggregation and duplication errors that are otherwise endemic in multi-source commodity data pipelines.

B.12 Junction Nodes, Collapsed States, and Latent Allocation

The crude-oil network contains many physical points where flows commingle or split: gathering manifolds where crude from multiple wells combines before entering the long-haul system, pump stations where a pipeline picks up or delivers volume at intermediate points, junctions where one pipeline connects to another, tank farms that act as pass-through buffers between producers and pipelines. These are real physical assets with real identities, and in the asset graph they are represented as nodes.

At most practical data resolutions, however, the variables that would distinguish a junction node from its immediate neighbours are not separately observable. If the only data available is basin-level production and terminal-level receipts, there is no independent observation of what a gathering manifold between the two is holding or moving. Any values placed on the junction node would be derived from the upstream and downstream aggregates — carrying no independent information, and at worst introducing spurious precision.

The framework handles this through the **collapsed state**. A junction node marked collapsed remains structurally present in the asset graph, but its flow variables are defined as pass-through formulas from its upstream neighbours to its downstream neighbours, its stock is held at zero, and its configuration features are retained but do not participate in the GNN's neighbourhood aggregation. In effect the node becomes a transparent conduit at the current resolution. When finer data later becomes available — for example, station-level pipeline telemetry from a commercial provider — the coverage contract is updated, the junction node's collapsed flag is removed, and it wakes up with its own observed flow variables. The topology is never rebuilt; only the scenario graph's interpretation of the existing topology changes.

Collapse is triggered in two distinct circumstances. The first, already discussed in Section B.10, is when data resolution is coarser than the node's natural level — a basin with only PADD-level data available is collapsed for that scenario. The second, specific to junction nodes, is when the node's role in the physical network is one of pure transit between observable neighbours, with no independent variables that the current data can populate. In both cases the mechanism is the same: the node is present in the asset graph, inactive in the scenario graph, and ready to activate when appropriate data becomes available.

A distinct problem arises at junctions where flows split across multiple downstream routes. The Permian basin's production is known in aggregate; the individual pipelines leaving the Permian — Gray Oak, Cactus II, Midland-to-ECHO, Wink-to-Webster, Basin — have known nominal capacities but the actual split of Permian outflow across them in any given month is generally not reported in public data. The split is determined by a combination of pipeline tariffs, contractual commitments, operational constraints, and spot-market economics, none of which are directly observable at the aggregate data level.

This is the **latent allocation problem**. Its treatment is deferred to Chapter 5 on the forecasting framework, but its structural implications belong here. Because the split is latent, the framework does not attempt to infer it from rules; instead, it treats the per-edge flow variable at each junction as a learnable quantity, constrained by the mass balance (the edge flows must sum to the junction's observed outflow), by pipeline capacity ceilings (no edge flow may exceed the receiving pipeline's nominal capacity), and by any partial observations that may exist (pipeline-specific tariff filings, downstream terminal receipts that can be attributed to specific pipelines). This is a natural use case for attention-based temporal graph models: GAT and TGAT learn attention weights over neighbours conditional on node state, and the Permian-to-pipeline allocation problem is exactly of that form. The attention weights on the outbound edges from the Permian node, learned from historical co-movement of basin production and downstream terminal inflows, approximate the latent allocation.

Framing this in the annex, rather than only in the forecasting chapter, clarifies why the asset-centric graph is the right representation even when the edge-level flows cannot be directly observed. The graph structure encodes the set of physically possible flows; the observed data constrains the aggregate sums; and the GNN learns the distribution of flow across the possible routes. Without the asset-centric graph, there would be nothing for the attention mechanism to attend over, and the allocation would have to be handled outside the model through ad-hoc rules.

Figure B.8 — A junction node under two different coverage contracts. In the basin-only contract, the Permian gathering node is collapsed and flows pass through transparently from the basin to the downstream pipelines, with the split learned as a latent allocation. In the station-level contract, the gathering node is active and its per-edge outflows are observed directly from commercial telemetry.

B.13 Summary and Position in the Thesis

Sections B.1 through B.6 establish the asset-centric convention and the universal mass balance with a balancing item. Sections B.7 through B.12 extend that foundation with the machinery needed to operate the framework at production scale: the separation of a persistent asset graph from the scenario graphs derived from it, the distinction between geography metadata as labels on physical nodes and a resolution hierarchy as relationships between nodes, the observed/derived invariant that makes aggregation double-counting structurally impossible, the coverage contracts that allow administrative aggregates to be promoted to authoritative status when no finer data exists, and the collapsed-state mechanism for junction nodes whose variables cannot be distinguished at the current resolution.

These extensions are not incidental engineering. They are the methodological contributions that make the asset-centric framework a practical basis for forecasting in an environment where data sources are heterogeneous, partial, and time-varying. Published graph schemas for supply chains typically assume a single fixed resolution and break when data quality improves; the present framework absorbs granularity upgrades through the coverage contract without modification to the asset graph, and it treats mixed-resolution scenarios as a first-class case rather than an exception. The observed/derived invariant recasts the most common source of error in multi-source commodity data — aggregation double-counting — from a runtime validation concern to a schema-level property the data model cannot violate. The collapsed-state mechanism allows the asset graph to be defined once at the full level of physical detail, with scenario-specific inactivity handled at load time rather than through topology edits.

The framework is taken up in Chapter 5 on the forecasting framework, where the latent allocation problem introduced in Section B.12 becomes the central modelling challenge and the attention-based temporal graph models discussed in Section 4.4 become the methodological solution. The scenario graph emitted by the loader is the direct input to those models; the observed/derived invariant ensures that the inputs they receive are internally consistent; and the coverage contract mechanism is what allows the same model to be evaluated on historical periods with coarser data and recent periods with finer data, without retraining from scratch.

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Annex C: Resolver Dispatch Rule Reference

This annex provides a concise reference for the resolver's dispatch rules, intended as a lookup aid for readers working with the implementation. The narrative treatment of the resolver appears in Chapter 6; this annex strips out the discussion and presents the rules in compact form.

C.1 Rule priority and short-circuit behaviour

The resolver evaluates each variable through a fixed priority list. The first rule that successfully produces a value short-circuits the remaining rules. The unresolved fallback (Rule 9) should fire zero times in a healthy run; non-zero unresolved counts are bugs and require operator action.

| Priority | Rule | Trigger condition | Output source |

|—|—|—|—|

| 1 | Observed | `timeseries_id IS NOT NULL` | observed |

| 2 | Zero | `formula = '0'` | zero |

| 3 | Latent (with reverse-mirror promotion) | `formula = 'latent()'` | latent or reverse-mirror |

| 4 | Sum | `formula = 'sum'` | derived |

| 5 | Single-variable alias | `formula = bare_variable_id` | derived |

| 6 | Reverse mirror (standalone) | paired direction resolved, this direction has no own formula | derived |

| 7 | Closure formula | balancing item with mixed-type inputs in formula_inputs | derived |

| 8 | Arithmetic | formula matches signed-variable-references pattern | derived |

| 9 | Unresolved (fallback) | none of the above matched | unresolved |

C.2 Rule details

C.2.1 Rule 1: Observed

The variable carries `timeseries_id IS NOT NULL`. The rule performs a direct lookup against `timeseries_data` at the observation date. The lookup is a primary key access on `(timeseries_id, observation_date)`.

Output: `value = <time series value>`, `source = 'observed'`, `formula_used = <timeseries_id>`.

Failure mode: The time series has no value at the observation date. The current implementation records `value = NULL`, `source = 'observed'` with a NULL value as a transient state, which the L4 views flag as a partial coverage warning. Future revision will distinguish "observed but null" from "unresolved".

C.2.2 Rule 2: Structural zero

The variable carries `formula = '0'`. Used for pass-through node defaults inherited from `node_type_default_formulas`: pipeline P, C, ΔS, B; refinery outflow; gathering P (within-basin); and other physical structural zeros.

Output: `value = 0`, `source = 'zero'`, `formula_used = '0'`.

Failure mode: None. This rule cannot fail.

C.2.3 Rule 3: Latent with reverse-mirror promotion

The variable carries `formula = 'latent()'`. The rule first attempts a reverse-mirror promotion: if the paired direction of the variable (opposite variable type, swapped endpoints) has a resolved value (and is itself not plain latent), the rule borrows that value through the mirror identity.

Outputs depending on the promotion:

- Mirror succeeded: `value = <paired direction's value>`, `source = 'reverse-mirror'`, `formula_used = 'reverse_mirror_promotion'`.
- Mirror failed: `value = NULL`, `source = 'latent'`, `formula_used = 'latent()'`.

The mirror-promotion logic exists because Rule 3's natural priority position (before Rule 6) would otherwise short-circuit and prevent the standalone Rule 6 from firing for explicitly-latent variables.

Failure mode: None at the rule level. Downstream LP consumers must treat `source = 'latent'` rows as decision variables, not as constants.

C.2.4 Rule 4: Sum

The variable carries `formula = 'sum'`. The rule sums every value in `formula_inputs`; NULL inputs are skipped (treated as zero contribution).

Output: `value = Σ inputs`, `source = 'derived'`, `formula_used = 'sum'`.

Failure mode: All inputs are NULL. The rule produces `value = 0` rather than NULL, which is potentially misleading. Operators should check the `formula_inputs` structure when a sum variable resolves to zero unexpectedly.

The rule is the twelfth-pass collapse of three earlier labels (`sum_over_children`, `sum_over_outflows`, `sum_same_type`). The semantic role of any given sum is recoverable from the structure of `formula_inputs` and does not need to be encoded in the formula text.

C.2.5 Rule 5: Single-variable alias

The variable carries `formula = <bare_variable_id>` (a bare identifier matching exactly one entry in `formula_inputs`). The rule inherits the value of the named target.

Output: `value = <target value>`, `source = 'derived'`, `formula_used = 'alias(<bare_variable_id>)'`.

Failure mode: The target is not resolved at this point in the dispatch loop. This should not happen if the dependency DAG is built correctly; if it does, the variable falls through to Rule 9 (unresolved).

C.2.6 Rule 6: Reverse mirror (standalone)

The variable has no own resolution path (no time series, no formula matching Rules 1 through 5) and its paired direction has a resolved value. The rule borrows the paired direction's value.

Output: `value = <paired direction's value>`, `source = 'derived'`, `formula_used = 'reverse_mirror'`.

Failure mode: The paired direction is itself unresolved. The variable falls through to Rule 9.

C.2.7 Rule 7: Closure formula

The variable is a balancing item whose `formula_inputs` mix inventory, inflow, and outflow variables. The rule evaluates the closure equation:

$B = \Delta S - P + C - \Sigma F_{in} + \Sigma F_{out}$

Output: `value = <computed B>`, `source = 'derived'`, `formula_used = 'closure'`.

This rule is dormant in the starter scenario because B was promoted to time-series-observed at the PADD and USA aggregates in the sixth migration pass. The rule remains in the resolver as a fallback and as the engine behind the observed-vs-closure-B comparison view of Section 7.4.

C.2.8 Rule 8: Arithmetic combination

The variable carries `formula = <signed variable references>`, for example `padd2_view.P - bakken_nd.P - oklahoma.P`. The rule parses the signed terms via a regular expression that extracts (sign, variable_id) pairs and aborts if any term fails to match a variable in `formula_inputs`.

Output: `value = Σ (sign × variable value)`, `source = 'derived'`, `formula_used = <original formula string>`.

Failure mode: Parser fails to match the formula text. The variable falls through to Rule 9.

C.2.9 Rule 9: Unresolved (fallback)

None of the above rules matched.

Output: `value = NULL`, `source = 'unresolved'`, `formula_used = NULL`.

Operator action required. A non-zero unresolved count after a run indicates that either the assignment set references variables that do not exist, or the formula text matches no rule pattern. The recommended response is to inspect the audit row's dispatch counts, query `scenario_resolved_values WHERE source = 'unresolved'` to identify the failing variables, and either fix the assignment (correct the reference) or extend the resolver (add a rule for the new pattern). Silent treatment as latent is explicitly prohibited.

C.3 The dependency DAG

The resolver's `build_deps()` function produces a dictionary `deps[variable]` of upstream variable IDs that each variable depends on. The topological sort over this dictionary orders the variables so that each is resolved only after its dependencies.

Three sources contribute to the dependency dictionary:

| Source | Trigger | Dependency added |

|—|—|—|

| Formula inputs | Variable has a non-empty `formula_inputs` list | Every variable ID in the list that exists in the assignment set |

| Fan-out at same node | Variable's formula is a sum over outflows | Every outflow variable at the same node, except itself |

| Reverse mirror, hoisted | Variable is latent and paired direction is resolved | The paired direction's variable ID |

Two cycle gates protect the reverse-mirror addition:

| Gate | Condition |

|—|—|

| Paired side is non-latent | Without this, latent ↔ latent forms a two-node cycle |

| Paired side does not already alias from us | Without this, this side aliases from paired, paired aliases from this side → cycle |

A cycle detected by the topological sort raises a `CycleError` and halts the resolver. The audit row records the failure with a non-NULL `failed_at` timestamp.

C.4 Healthy-run signature

The dispatch counts in a healthy run on the starter scenario are stable across repeated runs:

| Rule | Expected count |

|—|—|

| observed | 80 |

| zero | 628 |

| latent | 652 |

| alias | 442 |

| reverse-mirror | 15 |

| arithmetic | 8 |

| sum | 5 |

| closure | 0 |

| unresolved | 0 |

| **Total** | **1,830** |

Deviations from this signature in any rule other than `observed` (which scales linearly with the count of bound time series) indicate a structural change. The dispatch counts are recorded in `scenario_resolver_runs.dispatch_counts` as JSONB and can be diffed across runs with standard SQL.

Annex D: Variable Catalogue and Starter Scenario Inventory

This annex provides a structured inventory of the variables in the starter scenario `starter_us_crude_2015_2025`, intended as a reference for readers working with the implementation. The annex is not exhaustive — the full assignment set carries 1,830 variables — but it is comprehensive enough to give a clear picture of how the framework's principles materialise in the actual data model.

D.1 Node inventory by class

| Class | Count | Examples |

|—|—|—|

| Production (physical) | 19 | bakken_nd, bakken_mt, permian_tx, permian_nm, eagle_ford_tx, oklahoma_conventional, california_conventional, wyoming, padd1_other through padd5_other, gulf_of_america |

| Refining (physical) | 24 (active; 115 total slots) | padd1_refinery_agg, padd2_refinery_agg through padd5_refinery_agg, 19 named refineries (Motiva Port Arthur, ExxonMobil Beaumont, Marathon Galveston Bay, BP Whiting, Marathon Detroit, Valero refineries, and others) |

| Storage (physical) | 10 | cushing_hub (with three operator sub-terminals: enterprise_cushing, plains_cushing, magellan_cushing), patoka_hub, houston_storage, st_james_terminal, loop_terminal, others |

| SPR (physical) | 4 | bayou_choctaw, big_hill, bryan_mound, west_hackberry |

| Pipeline (physical) | 37 | seaway_south, seaway_north, capline_south, capline_north, keystone_xl, marketlink, diamond, pipe_bakken_xstate, 12 inter_padd__*to*__agg aggregates, and others |

| Import terminal (physical) | 8 | clearbrook, padd1_imports_agg through padd5_imports_agg, padd1_canadian_imports_agg, padd2_canadian_imports_agg, padd2_xstate |

| Export terminal (physical) | 12 | ingleside, houston_exports_agg with three sub-terminals (enterprise_houston, magellan_houston, moda_midstream_houston), nederland, loop_exports, st_james_exports, padd1 through padd5 export aggregates |

| Gathering (physical) | 6 | ans, bakken_nd_gathering, bakken_mt_gathering, eagle_ford_gathering, permian_nm_gathering, permian_tx_gathering |

| Origin terminal (physical) | 6 | midland, wink, crane, johnsons_corner, three_rivers, valdez |

| Region view (abstract) | 6 | usa_view, padd1_view through padd5_view |

| Refining district view (abstract) | 10 | refining_district_1A, 1B, 2A, 2B, 2C, 3A, 3B, 3C, 4_west, 5_combined |

| State view (abstract) | 2 | texas_state_view, montana_state_view |

| Observational aggregate (abstract) | 4 | spr_total, permian_aggregate, bakken_aggregate, eagle_ford_aggregate |

| USA subtotal view (abstract) | 1 | usa_lower48_excl_gom_view |

| Boundary | 4 | foreign_supply, canadian_oil_sands, foreign_export_destination, spr_total (also boundary-functional) |

| **Total** | **240** | |

D.2 Variables by type

The starter scenario carries 1,830 variables, distributed across the seven variable types as follows:

| Type | Symbol | Count | Note |

|—|—|—|—|

| Production | P | 245 | One per (node × commodity), with non-applicable types set to zero |

| Consumption | C | 245 | Same structure as P |

| Inventory | S | 245 | Same structure as P |

| Balancing item | B | 245 | Same structure as P |

| Inflow | F_in | 410 | One per directed inbound edge per node |

| Outflow | F_out | 410 | One per directed outbound edge per node (matches F_in count) |

| Phi (reference) | φ | 30 | Used for cross-node alias chains in derived aggregates |

| **Total** | | **1,830** | |

The 245-vs-240 difference for the non-relational types arises because three nodes have multiple commodity bindings (planned future-work decomposition into light sweet and heavy sour grades; not yet active in the resolver, but the variables exist as latent placeholders).

D.3 Authoritative bindings by subsystem

The 80 time-series-observed variables in the starter scenario are distributed across the U.S. crude system as follows:

| Subsystem | Variable type | Authoritative resolution | Count of bound variables | Lead EIA series |

|—|—|—|—|—|

| Production | P | Basin level + state | 15 | MCRFPP1, MCRFPP2, MCRFPP3, MCRFPP4, MCRFPP5, MCRFPP3TX, MCRFPP3NM, others |

| Refinery throughput | C | PADD level | 5 | MCRRIP1 through MCRRIP5 |

| Stocks (commercial) | S | PADD level | 5 | MCESTP1 through MCESTP5 |

| Stocks (SPR sites) | S | Site level | 4 | MCESTUS1 series (decomposed) |

| Stocks (Cushing hub) | S | Hub level | 1 | Cushing weekly aggregated to monthly |

| Imports (non-Canada) | F_in | PADD level | 5 | MCRIPP12 minus Canadian contribution |

| Imports (Canada) | F_in | PADD level | 2 | MCRIPP1CA2 (PADD 1 Canadian), MCRIPP2CA2 (PADD 2 Canadian) |

| Exports | F_out | PADD level | 5 | MCREXP12 through MCREXP52 |

| Inter-PADD flows | F (aggregate) | PADD-pair level | 12 | MCRMP* family across all pairs |

| Balancing item | B | PADD level + USA | 6 | MCRUA-P1 through MCRUA-P5 and MCRUA-USA |

| USA aggregates | P, C, S, F_in, F_out, B | USA total | 6 | MCRFPUS2, MCRRIUS2, MCESTUS1, MCRIMUS2, MCREEXUS2, MCRUA |

| Texas/Montana state aggregates | P | State | 2 | Texas state series, Montana state series |

| Auxiliary observed (not in mass balance) | S | PADD level | 12 | MCRSFP (PADD ex-SPR), MCRSSP (SPR), others |

| **Total** | | | **80** | |

The 80 authoritative bindings collectively produce all 1,830 resolved values through the resolver's dispatch logic.

D.4 Latent variables by location

The 652 latent variables in the starter scenario are concentrated at junction nodes where per-edge flows are not publicly observed. The distribution:

| Junction or subsystem | Latent count | Reason |

|—|—|—|

| Permian fan-out (basin → origin terminals → trunk pipelines) | 47 | Per-pipeline splits between Midland, Wink, Crane terminals and downstream Gulf Coast pipelines |

| Bakken fan-out (gathering → trunk pipelines to PADD 2 and beyond) | 32 | Per-pipeline splits out of Bakken gathering systems |

| Bakken cross-state connector | 4 | All four flow variables on `pipe_bakken_xstate` (eleventh migration pass) |

| Cushing hub sub-decomposition | 18 | Per-operator-sub-terminal splits within the hub (only aggregate stocks observed) |

| Houston exports sub-decomposition | 24 | Per-sub-terminal splits within the Houston aggregate |

| Inter-PADD aggregate per-pipeline splits | 196 | Each inter-PADD aggregate has multiple constituent pipelines; aggregate flow observed, per-pipeline split latent |

| SPR per-site outflows | 16 | Observed only at SPR-total level when released; per-site decomposition latent |

| Refinery per-pipeline-receipt flows | 184 | Refineries observed at PADD-aggregate level for receipts; per-source-pipeline splits latent |

| Origin terminal outflows to multiple downstream pipelines | 89 | Midland, Wink, Crane outflows to multiple trunk pipelines, latent at the per-pipeline level |

| Offshore production routing | 28 | Gulf of America platforms route through multiple offshore-to-shore systems, splits unobserved |

| Other (smaller subsystems) | 14 | Foreign aggregate boundary edges, miscellaneous latents |

| **Total** | **652** | |

These latents are the natural decision variables for the linear programme described in Chapter 8. Each carries a structural relationship to the variables around it via mass balance and capacity constraints, but its value is not pinned down by observation.

D.5 Structural zeros by node type

The 628 structural zero variables are inherited from `node_type_default_formulas`. The distribution by node type:

| Node type | Structural zeros per node | Total nodes | Zero variables |

|—|—|—|—|

| Pipeline | 4 (P, C, ΔS, B; the four pass-through quantities) | 37 | 148 |

| Gathering | 4 (P, C, ΔS, B if no own production) + 1 (P for gatherings within their basin: zero if basin-level P is authoritative elsewhere) | 6 | 24 |

| Refinery | 1 (P; refineries do not produce crude) | 24 | 24 |

| Storage terminal | 4 (P, C, ΔS for pass-through stocks, B) | 10 | 40 |

| SPR site | 2 (P, C; SPR sites do not produce or consume) | 4 | 8 |

| Import terminal | 4 (P, C, ΔS for pass-through, B) | 8 | 32 |

| Export terminal | 4 (P, C, ΔS for pass-through, B) | 12 | 48 |

| Origin terminal | 4 (P, C, ΔS, B for pass-through) | 6 | 24 |

| Boundary source | C, ΔS, B | 3 (foreign_supply, canadian_oil_sands, plus structural slots) | 9 |

| Other (region views, observational, mixed) | various | 130 | 271 |

| **Total** | | | **628** |

The total of 628 matches the dispatch count of `zero` rule firings in the starter scenario.

D.6 Migration history

The starter graph reached its current state through twelve migration passes. The major passes are summarised here:

| Pass | Description | Result on dispatch counts |

|—|—|—|

| 1 | Initial seed load from `asset_graph.json` | Baseline ~1,200 variables, ~50% unresolved |

| 2 | Populate `node_type_default_formulas`; resolve pass-through zeros | Unresolved drops to ~30% |

| 3 | Add PADD-level residual nodes; assign EIA PADD aggregates | Unresolved drops to ~20%; arithmetic rule first appears |

| 4 | Wire inter-PADD aggregates; add the MCRMP__ series | Aliases proliferate; reverse-mirror first fires (3 cases) |

| 5 | Add SPR sub-nodes; promote SPR site stocks to authoritative | SPR-aggregate becomes derived; 4 new observed bindings |

| 6 | Promote balancing item B to time-series-observed at PADD and USA | Closure rule fires drop from ~15 to 0 |

| 7 | Add origin terminal nodes (Midland, Wink, Crane); declare Permian fan-out latents | Latent count rises to ~400 |

| 8 | Fix latent-mirror cycle; add cycle gates to `build_deps()` | Resolver runs to completion without `CycleError` |

| 9 | Hoist reverse-mirror dependency above early-continue; promote mirror inside Rule 3 | Reverse-mirror dispatch lifts from 3 to 15 |

| 10 | Add Bakken cross-state connector node; declare its four flow latents | Latent count rises to ~640 |

| 11 | Sub-decompose Houston exports; reorganise Cushing hub sub-terminals | Latent count reaches 652 |

| 12 | Collapse three sum labels (`sum_over_children`, `sum_over_outflows`, `sum_same_type`) into canonical `sum` | Dispatch surface reduced from 11 labels to 9 |

Each migration pass is implemented as a Python script in the `migrations/` directory and is replayable from a fresh database. The current state of the starter scenario is the result of running all twelve passes in sequence.

D.7 Summary statistics

For reference at a glance:

- **Nodes:** 240 (197 physical, 43 abstract; including 4 boundary)
- **Edges:** 409 directed flow edges
- **Variables:** 1,830 under the active scenario
- **Time-series bindings:** 80 (in 1:1 attribution)
- **Observation dates:** 156 (monthly, January 2015 to December 2024)
- **Resolved (variable, date) pairs:** 285,480
- **Resolver runtime:** approximately 20 seconds on a developer laptop
- **Partition closure gap:** 0 kbd at every PADD aggregate and at the USA aggregate
- **Unresolved variables:** 0 in the current healthy state

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